

Contents

1	Smooth de Rham cohomology	2
1.1	The complex and its quotient spaces	2
1.2	Poincaré lemma and an explicit homotopy operator	3
1.3	A resolution and the comparison with singular cohomology	4
1.4	The punctured plane: a period detects the hole	6
1.5	The square torus and the absence of a preferred representative	7
1.6	Orientation, volume forms, and Poincaré duality	7
2	A metric on forms and the Hodge star	9
2.1	Pointwise inner products come first	9
2.2	Definition and algebra of the star	9
2.3	From pointwise lengths to the global L^2 inner product	10
2.4	Flat tori: lengths, energies, and the choice of lattice	11
2.5	Calculations in the plane and in three dimensions	12
2.6	Four dimensions: self-dual and anti-self-dual two-forms	13
2.7	The round sphere: local calculations and global forms	15
2.8	The hyperbolic upper half-plane and quotients	15
3	The adjoint, the Laplacian, and the analytic bridge	16
3.1	Deriving the codifferential from Stokes' theorem	16
3.2	The Laplacian and what follows without analysis	17
3.3	A complete finite dimensional Hodge theorem	18
3.4	Why infinite dimension requires a new theorem	19
3.5	The flat torus: Fourier analysis makes the bridge explicit	20
3.6	Poisson's equation on the torus: obstruction, inverse, estimates	22
3.7	Ellipticity: the highest order part is invertible	24
3.8	The analytic black box, and what we prove from it	24
4	Hodge decomposition, duality, and complete examples	26
4.1	From the Green operator to the decomposition	26
4.2	Poincaré duality from a positive integral	28
4.3	The round two-sphere: derivatives, eigenforms, and duality	28
4.4	The flat torus: harmonic representatives and all three summands	29
4.5	Flat metrics, periods, and the four-dimensional pairing	31
4.6	Changing the metric: a surface phenomenon	32
4.7	The hyperbolic upper half-plane and a compact quotient	34
5	Exercises and hints	36

1 Smooth de Rham cohomology

Throughout this topic, a smooth manifold is finite-dimensional, Hausdorff, second countable, and without boundary unless explicitly stated otherwise. In particular, it is paracompact and admits smooth partitions of unity. The word *closed*, when applied to a manifold, means compact without boundary. Connectedness is not assumed. Forms have real coefficients unless stated otherwise; smooth forms are denoted by $\mathcal{A}^k(M)$.

Where compactness enters. The smooth de Rham complex, the Poincaré lemma, the fine resolution, and the comparison with singular cohomology require neither compactness nor an orientation or a metric. Pointwise metric and star formulas are also independent of compactness. A global L^2 pairing requires square integrability; compactness makes every smooth form square-integrable. The global Hodge and Green-operator theorems proved below assume a compact Riemannian manifold without boundary. We use an orientation to express the star as an operator on ordinary forms and to write integration pairings; the Hodge theorem itself can also be formulated on a nonorientable closed manifold using the Riemannian density. Within these compact theorems, connectedness is needed only for claims that harmonic functions, or harmonic top forms, have one scalar parameter over the whole manifold.

1.1 The complex and its quotient spaces

For an n -manifold M , put

$$\mathcal{A}^k(M) = \Gamma(M, \Lambda^k T^*M), \quad \mathcal{A}^k(M) = 0 \quad \text{if } k < 0 \text{ or } k > n.$$

At a point p , a k -form is an alternating k -linear function on $T_p M$; a smooth form is a smoothly varying family of such functions. In coordinates $x^1, \dots, x^n$, write

$$\alpha = \sum_{i_1 < \dots < i_k} \alpha_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

The exterior derivative is given by

$$d\alpha = \sum_{i_1 < \dots < i_k} \sum_{j=1}^n \frac{\partial \alpha_{i_1 \dots i_k}}{\partial x^j} dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_k}. \quad (1.1)$$

It satisfies $d^2 = 0$ and the graded Leibniz rule

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta, \quad \alpha \in \mathcal{A}^k(M).$$

For every smooth map $F : M \rightarrow N$, one has $dF^* = F^*d$. These identities give the *de Rham complex*

$$0 \longrightarrow \mathcal{A}^0(M) \xrightarrow{d} \mathcal{A}^1(M) \xrightarrow{d} \dots \xrightarrow{d} \mathcal{A}^n(M) \longrightarrow 0. \quad (1.2)$$

We denote this smooth complex by $(\mathcal{A}^\bullet(M), d)$, reserving Ω for holomorphic or algebraic differential forms in later lectures. Its terms are global smooth sections. The corresponding sheaf of smooth k -forms is $\mathcal{A}_M^k$, with $\Gamma(M, \mathcal{A}_M^k) = \mathcal{A}^k(M)$.

Definition 1.1. A form is *closed* if $d\alpha = 0$ and *exact* if $\alpha = d\beta$. The k th de Rham cohomology is

$$H_{\text{dR}}^k(M) = \frac{\ker(d : \mathcal{A}^k(M) \rightarrow \mathcal{A}^{k+1}(M))}{\text{im}(d : \mathcal{A}^{k-1}(M) \rightarrow \mathcal{A}^k(M))}.$$

Thus $[\alpha] = [\alpha']$ precisely when $\alpha' - \alpha = d\beta$. In degree zero, $H_{\text{dR}}^0(M)$ consists of locally constant functions.

The quotient remembers what cannot be changed by adding an exact form. It does not, by itself, specify a representative. The possible representatives of $[\alpha]$ form the affine space $\alpha + d\mathcal{A}^{k-1}(M)$. The wedge product induces a graded-commutative product

$$[\alpha][\beta] = [\alpha \wedge \beta], \quad [\alpha][\beta] = (-1)^{k\ell}[\beta][\alpha].$$

For example, if $d\beta = 0$, then $(\alpha + d\gamma) \wedge \beta - \alpha \wedge \beta = d(\gamma \wedge \beta)$. This verifies independence of the first representative; the other factor is handled in the same way.

Example 1.2 (Two coordinate calculations). On $\mathbb{R}^2$, if $\alpha = P dx + Q dy$, then

$$d\alpha = (Q_x - P_y) dx \wedge dy.$$

Thus closedness is the integrability condition $Q_x = P_y$ for $d\alpha = 0$. On $\mathbb{R}^3$,

$$d(xz dy + y^2 dz) = z dx \wedge dy + (2y - x) dy \wedge dz.$$

These calculations require a smooth structure and coordinates, but no metric.

1.2 Poincaré lemma and an explicit homotopy operator

Proposition 1.3 (Homotopy formula). *Let $F : [0, 1] \times M \rightarrow N$ be a smooth homotopy and $f_t(x) = F(t, x)$. There is a degree -1 operator $K : \mathcal{A}^k(N) \rightarrow \mathcal{A}^{k-1}(M)$ such that*

$$f_1^* - f_0^* = dK + Kd. \quad (1.3)$$

Proof. Decompose the pullback uniquely as

$$F^*\alpha = \alpha_t + dt \wedge \beta_t, \quad K\alpha = \int_0^1 \beta_t dt.$$

Here α_t and β_t are forms on M of degrees k and $k-1$, respectively. The product differential gives

$$dF^*\alpha = d_M\alpha_t + dt \wedge (\partial_t\alpha_t - d_M\beta_t).$$

Since $F^*d\alpha = dF^*\alpha$, integrating its dt coefficient gives

$$Kd\alpha = \alpha_1 - \alpha_0 - d_M \int_0^1 \beta_t dt,$$

which is the claimed identity. For $k = 0$, set $K\alpha = 0$; the same calculation reads $Kd\alpha = f_1^*\alpha - f_0^*\alpha$. $\square$

Theorem 1.4 (Poincaré lemma). *If $U \subset \mathbb{R}^n$ is open and star-shaped about the origin, then $H_{\text{dR}}^0(U) = \mathbb{R}$ and $H_{\text{dR}}^k(U) = 0$ for $k > 0$. In particular, every closed positive-degree form is locally exact on any smooth manifold.*

Proof. Use $F(t, x) = tx$. The map f_1 is the identity and f_0 is constant. The pullback by f_0 vanishes in positive degree, so for a closed k -form with $k > 0$, equation (1.3) gives $\alpha = dK\alpha$. For a function f , the identity instead reads $f - f(0) = Kdf$. Consequently $df = 0$ implies that f is constant, which proves degree zero. Every point has a coordinate neighborhood corresponding to a ball, so the same argument gives the local assertion. $\square$

For actual calculations, the radial operator can be written without any notation about pullbacks:

$$(K\alpha)_x(v_1, \dots, v_{k-1}) = \int_0^1 t^{k-1} \alpha_{tx}(x, v_1, \dots, v_{k-1}) dt, \quad k \geq 1. \quad (1.4)$$

The vectors on the right are identified by the usual trivialization of $T\mathbb{R}^n$. In particular, for $\alpha = \sum a_j(x) dx^j$,

$$K\alpha(x) = \int_0^1 \sum_j x^j a_j(tx) dt.$$

For $\alpha = y dx + x dy$, this gives $K\alpha = \int_0^1 2txy dt = xy$, and indeed $d(xy) = \alpha$. For the constant two-form $dx \wedge dy$, it gives $K(dx \wedge dy) = \frac{1}{2}(x dy - y dx)$, whose differential is $dx \wedge dy$.

Equation (1.3) also shows that smoothly homotopic maps induce the same map on de Rham cohomology: on a closed form their pullbacks differ by $dK\alpha$. We will use this observation to reduce the punctured plane to a circle.

Example 1.5 (Local primitives on a torus). Let $q : \mathbb{R}^2 \rightarrow T^2 = \mathbb{R}^2/\mathbb{Z}^2$ be the quotient map. Every sufficiently small coordinate disk $U \subset T^2$ has a lift $s_U : U \rightarrow \mathbb{R}^2$. The functions $x_U = x \circ s_U$ and $y_U = y \circ s_U$ are well-defined on U , and the global forms denoted by dx, dy restrict to dx_U, dy_U . Thus dx has a primitive in every such disk. On each connected component of an overlap, two lifts differ by a lattice translation, so

$$x_U - x_V = m_{UV}, \quad y_U - y_V = n_{UV}, \quad m_{UV}, n_{UV} \in \mathbb{Z}.$$

These locally constant differences explain how derivatives can agree even when the primitives do not. Continuing a local primitive for dx once around the horizontal circle increases it by 1; it cannot be a global real-valued function. Its primitive upstairs is the coordinate function x , which is smooth on $\mathbb{R}^2$ but is not periodic.

The same issue occurs in degree two. Upstairs, $dx \wedge dy = d(x dy)$, but $x dy$ changes by dy under the horizontal translation. Local exactness is a statement about sufficiently small charts, and does not require these primitives to be periodic. This is the precise distinction that the sheaf resolution below will encode.

1.3 A resolution and the comparison with singular cohomology

Let $\underline{\mathbb{R}}_M$ denote the sheaf of locally constant real functions, and let $\mathcal{A}_M^k$ denote the sheaf $U \mapsto \mathcal{A}^k(U)$. The distinction between a sheaf and its global sections matters here: local primitives need not patch to a global primitive. The Poincaré lemma is exactly the assertion that

$$0 \longrightarrow \underline{\mathbb{R}}_M \longrightarrow \mathcal{A}_M^0 \xrightarrow{d} \mathcal{A}_M^1 \xrightarrow{d} \dots \xrightarrow{d} \mathcal{A}_M^n \longrightarrow 0 \quad (1.5)$$

is an exact sequence of sheaves: exactness is checked on germs near each point. Such an exact sequence is called a *resolution* of $\underline{\mathbb{R}}_M$.

The sheaves $\mathcal{A}_M^k$ are *fine*. Concretely, for any locally finite open cover, multiplication by a subordinate smooth partition of unity gives sheaf endomorphisms supported in the respective open sets, whose sum is the identity. Fine sheaves on a paracompact Hausdorff space are acyclic for global sections:

$$H^q(M, \mathcal{A}_M^k) = 0 \quad (q > 0).$$

This is the partition-of-unity input that makes the following homological algebra theorem applicable.

Remark 1.6 (Why a partition of unity does not remove the torus obstruction). In Example 1.5, let (ρ_U) be a smooth partition of unity subordinate to the coordinate disks. The locally defined products $\rho_U x_U$, extended by zero, give a global function $f = \sum_U \rho_U x_U$. Its derivative is

$$df = \sum_U \rho_U dx_U + \sum_U x_U d\rho_U = dx + \sum_U x_U d\rho_U.$$

The additional term need not vanish: although $\sum_U d\rho_U = 0$, the different x_U are not one common function. If a denotes the positive horizontal circle, the extra term has period -1 around a , since $\int_a df = 0$ and $\int_a dx = 1$. Fine sheaves therefore let us compute global cohomology from the resolution; they do not make the global de Rham complex exact. The nonzero class of dx is the obstruction to patching its local primitives.

Theorem 1.7 (Acyclic resolution theorem). *Let $\mathcal{F}$ be a sheaf of real vector spaces on a space X . Suppose $0 \rightarrow \mathcal{F} \rightarrow \mathcal{E}^0 \rightarrow \mathcal{E}^1 \rightarrow \dots$ is exact and $H^q(X, \mathcal{E}^j) = 0$ for every $q > 0$ and $j \geq 0$. Then there are canonical isomorphisms*

$$H^k(X, \mathcal{F}) \cong H^k(\Gamma(X, \mathcal{E}^\bullet)).$$

Here sheaf cohomology means the right derived functors of $\Gamma(X, -)$. This is the sheaf form of Leray's acyclicity lemma [1, Lemma 13.16.7, Tag 015E].

One can see the mechanism by putting $\mathcal{Z}^0 = \mathcal{F}$ and $\mathcal{Z}^j = \ker(\mathcal{E}^j \rightarrow \mathcal{E}^{j+1})$ for $j \geq 1$. The short exact sequences $0 \rightarrow \mathcal{Z}^j \rightarrow \mathcal{E}^j \rightarrow \mathcal{Z}^{j+1} \rightarrow 0$ give, for $k \geq 1$,

$$\frac{\Gamma(X, \mathcal{Z}^k)}{\text{im } \Gamma(X, \mathcal{E}^{k-1})} \cong H^1(X, \mathcal{Z}^{k-1}) \cong \dots \cong H^k(X, \mathcal{F}).$$

The first quotient is precisely the cohomology of the global complex. Applying the theorem to (1.5) therefore gives

$$H_{\text{dR}}^k(M) \cong H^k(M, \mathbb{R}_M). \quad (1.6)$$

There is a second standard comparison theorem: on a paracompact, locally contractible Hausdorff space, sheaf cohomology of the constant sheaf agrees with singular cohomology. A smooth manifold satisfies these hypotheses. Together the two comparisons give the de Rham isomorphism

$$H_{\text{dR}}^k(M) \cong H_{\text{sing}}^k(M; \mathbb{R}). \quad (1.7)$$

For this comparison and its identification with integration, see [2, Chapter 5]. We now make the map explicit.

Let $C_k^\infty(M; \mathbb{R})$ be the real vector space generated by smooth singular simplices $\sigma : \Delta^k \rightarrow M$; smoothness means smoothness up to the boundary, locally extending across the faces. Define

$$I : \mathcal{A}^k(M) \longrightarrow C_\infty^k(M; \mathbb{R}), \quad I(\alpha)(\sigma) = \int_{\Delta^k} \sigma^* \alpha.$$

For $k = 0$, this is evaluation at the point σ . With the singular coboundary convention $(\delta c)(\sigma) = c(\partial\sigma)$, Stokes' theorem gives the calculation

$$I(d\alpha)(\sigma) = \int_{\Delta^{k+1}} d\sigma^* \alpha = \int_{\partial\Delta^{k+1}} \sigma^* \alpha = \delta I(\alpha)(\sigma). \quad (1.8)$$

Thus integration is a cochain map. In particular, the period $\int_c \alpha$ of a closed form depends only on $[\alpha]$ and the smooth singular homology class $[c]$.

The smoothing comparison theorem says that the inclusion of smooth singular chains into continuous singular chains induces an isomorphism on homology and, by restriction, on real cohomology. Integration therefore yields

$$H_{\text{dR}}^k(M) \xrightarrow{I} H^k(C_\infty^\bullet(M; \mathbb{R})) \xleftarrow[\cong]{\text{restriction}} H_{\text{sing}}^k(M; \mathbb{R}).$$

The de Rham theorem asserts that the left arrow is also an isomorphism; the resulting map to ordinary singular cohomology is precisely (1.7). In a sheaf proof, integration is compatible with restriction and induces the identity on locally constant functions in degree zero. Comparison of resolutions then identifies it with the canonical map above. A full proof uses a suitable sheaf resolution of singular cochains; the assignment of ordinary singular cochains to open sets is not itself a sheaf.

Remark 1.8. Integration is canonical and natural under smooth pullback. The induced isomorphism is an isomorphism of graded rings, matching the wedge product with the singular cup product. This multiplicativity is an assertion on cohomology; the usual simplex integration map does not in general take a wedge product to the usual cup product as an equality of cochains. No compactness, orientation on M , or metric is used in these comparisons: integration takes place over compact oriented simplices, even when M is noncompact or nonorientable. The comparison does not imply that the cohomology of a noncompact manifold is finite dimensional.

1.4 The punctured plane: a period detects the hole

On $U = \mathbb{R}^2 \setminus \{0\}$, define the normalized angular form

$$\eta = \frac{1}{2\pi} \frac{-y dx + x dy}{x^2 + y^2}. \quad (1.9)$$

Writing the unnormalized coefficients as $P = -y/(x^2 + y^2)$ and $Q = x/(x^2 + y^2)$, one obtains

$$Q_x = \frac{y^2 - x^2}{(x^2 + y^2)^2} = P_y, \quad d\eta = 0.$$

For the positively oriented unit circle $\gamma(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$,

$$\gamma^* \eta = \frac{dt}{2\pi}, \quad \int_\gamma \eta = 1.$$

An exact form integrates to zero on a closed curve, so η is not exact. Locally it is $d\theta/(2\pi)$; a single-valued global angle would contradict its nonzero period.

For the complete calculation, the smooth homotopy $F(t, x) = ((1-t) + t/|x|)x$ retracts U onto S^1 . On the circle a one-form pulls back to $a(t) dt$ with a a 2π -periodic function. If $c = \int_0^{2\pi} a(t) dt$, then

$$f(t) = \int_0^t \left(a(s) - \frac{c}{2\pi} \right) ds$$

is periodic and $a(t) dt = c dt/(2\pi) + df$. Consequently

$$H_{\text{dR}}^k(U) = \begin{cases} \mathbb{R}[1], & k = 0, \\ \mathbb{R}[\eta], & k = 1, \\ 0, & k \geq 2. \end{cases} \quad (1.10)$$

Every closed one-form on U is $c\eta + df$, with $c = \int_\gamma \alpha$; the function f is unique up to a constant. The coordinate expression for η is useful, but smooth topology alone does not privilege it over $\eta + df$.

1.5 The square torus and the absence of a preferred representative

Write

$$T^2 = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z}) = \mathbb{R}^2/\mathbb{Z}^2, \quad z = x + iy.$$

Thus the notation $\mathbb{C}/\mathbb{Z}^2$ means that the abstract group $\mathbb{Z}^2$ has been embedded as the lattice generated by 1 and i . The functions x, y do not descend to real-valued functions on T^2 , whereas dx, dy do, since they are invariant under lattice translations.

Let $a(t) = (t, 0)$ and $b(t) = (0, t)$, with $0 \leq t \leq 1$, denote the two fundamental loops. If α is closed, put $A = \int_a \alpha$ and $B = \int_b \alpha$. Its pullback to $\mathbb{R}^2$ has a primitive F by the Poincaré lemma. Since dF is periodic, each difference $F(x+1, y) - F(x, y)$ and $F(x, y+1) - F(x, y)$ is constant; evaluating on the fundamental loops gives A and B , respectively. Hence $f = F - Ax - By$ is periodic and descends to the torus. This proves the explicit normal form

$$\alpha = A dx + B dy + df. \quad (1.11)$$

The period matrix of dx, dy on a, b is the identity, so these two classes are independent.

There is an equally explicit calculation in degree two. For a periodic function $h(x, y)$, set

$$c = \int_0^1 \int_0^1 h(x, y) dx dy, \quad m(y) = \int_0^1 h(x, y) dx,$$

$$P(x, y) = \int_0^x (h(s, y) - m(y)) ds, \quad Q(y) = \int_0^y (m(t) - c) dt.$$

Both P and Q are periodic, and direct differentiation gives

$$h dx \wedge dy = c dx \wedge dy + d(P dy - Q dx).$$

Since $\int_{T^2} dx \wedge dy = 1$, the constant term is exact only if $c = 0$. We have therefore computed the entire ring:

$$H_{\text{dR}}^\bullet(T^2) = \Lambda^\bullet \langle [dx], [dy] \rangle, \quad (b_0, b_1, b_2) = (1, 2, 1). \quad (1.12)$$

For instance, $dx + d(\sin(2\pi x)) = (1 + 2\pi \cos(2\pi x)) dx$ has exactly the same periods and class as dx .

Remark 1.9 (What “no canonical representative” means). The lattice presentation supplies an affine structure, which singles out constant-coefficient forms. This is extra structure beyond the underlying smooth manifold. Indeed, suppose a representative of $[dx]$ were invariant under all diffeomorphisms isotopic to the identity. Invariance under all torus translations would force its coefficients to be constant, and its periods would force it to equal dx . But the diffeomorphism

$$\phi(x, y) = (x + \varepsilon \sin(2\pi x), y), \quad 0 < |\varepsilon| < \frac{1}{2\pi},$$

is isotopic to the identity and satisfies $\phi^* dx = dx + d(\varepsilon \sin(2\pi x)) \neq dx$. Thus there is no such diffeomorphism-natural representative of this class from the smooth structure alone. On a compact manifold without boundary, a Riemannian metric will select a unique harmonic representative.

1.6 Orientation, volume forms, and Poincaré duality

An orientation specifies which ordered bases of $T_p M$ are positive. On an oriented n -manifold, a *positive volume form* is a nowhere-zero n -form μ that evaluates positively on every positive basis. Such a form can be chosen without a metric, but an orientation does not determine one: if $f > 0$

is smooth, then $f\mu$ is another choice. For a compact oriented manifold without boundary, Stokes' theorem gives

$$\int_M d\gamma = 0, \quad \int_M : H_{\text{dR}}^n(M) \longrightarrow \mathbb{R}.$$

If M is connected, this last map is an isomorphism. In particular, $\mu / \int_M \mu$ represents the uniquely normalized top-degree class, although its differential-form representative still depends on μ . If M has several connected components, integrate over each component: this identifies $H_{\text{dR}}^n(M)$ with one copy of $\mathbb{R}$ per component. The total integral alone is then not injective. Compactness makes the number of components finite.

Example 1.10 (Changing a volume form on the torus). On the oriented square torus, take

$$\mu_\varepsilon = (1 + \varepsilon \cos(2\pi x)) dx \wedge dy, \quad |\varepsilon| < 1.$$

This is a positive volume form with total volume 1, and

$$\mu_\varepsilon - dx \wedge dy = d\left(\frac{\varepsilon}{2\pi} \sin(2\pi x) dy\right).$$

Thus all these forms represent the same normalized top-degree class. The primitive is now periodic, unlike $x dy$ in Example 1.5. The zero integral of the difference is exactly the solvability condition found in the explicit degree-two computation above. After a metric is chosen, harmonicity will select a particular representative of this normalized class.

Example 1.11 (The sphere before introducing a metric). Orient $S^2 \subset \mathbb{R}^3$ by its outward normal. The restriction

$$\omega = (x dy \wedge dz + y dz \wedge dx + z dx \wedge dy)|_{S^2}$$

is positive: at $p \in S^2$, $\omega_p(v, w) = \det(p, v, w)$. Using $p = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$, one calculates

$$\omega = \sin \theta d\theta \wedge d\varphi, \quad \int_{S^2} \omega = \int_0^{2\pi} \int_0^\pi \sin \theta d\theta d\varphi = 4\pi.$$

The Cartesian expression is smooth at the poles, where these coordinates break down. By the de Rham theorem and the usual cohomology of S^2 , $H_{\text{dR}}^1(S^2) = 0$ and $H_{\text{dR}}^2(S^2) = \mathbb{R}[\omega/(4\pi)]$. Thus every closed one-form on S^2 is exact, and a two-form is exact precisely when its integral is zero.

Theorem 1.12 (Poincaré duality in de Rham form). *Let M be a compact oriented n -manifold without boundary. The pairing*

$$H_{\text{dR}}^k(M) \times H_{\text{dR}}^{n-k}(M) \longrightarrow \mathbb{R}, \quad ([\alpha], [\beta]) \longmapsto \int_M \alpha \wedge \beta \quad (1.13)$$

is nondegenerate. Equivalently, it identifies $H_{\text{dR}}^k(M)$ with the dual vector space of $H_{\text{dR}}^{n-k}(M)$.

Well-definedness follows immediately from the Leibniz rule and Stokes: if $d\beta = 0$, then $\int_M d\gamma \wedge \beta = \int_M d(\gamma \wedge \beta) = 0$. The nondegeneracy is the substantive theorem; later in this topic the Hodge star will give a short proof once harmonic representatives exist. On the oriented square torus it is already explicit:

$$\int_{T^2} (A dx + B dy) \wedge (C dx + D dy) = AD - BC.$$

The matrix in the basis $[dx], [dy]$ is $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, which is invertible. On S^2 , the classes $[1]$ and $[\omega/(4\pi)]$ pair to 1.

Remark 1.13 (Why compactness appears). The punctured plane has $H^0 = \mathbb{R}$ but $H^2 = 0$, so the displayed ordinary cohomology pairing cannot be nondegenerate there. For an oriented manifold without boundary that is not compact, the appropriate version pairs $H_{\text{dR}}^k(M)$ with compactly supported cohomology $H_{\text{dR},c}^{n-k}(M)$ by the same integral. The compact support makes the integral finite and permits Stokes' theorem. Here $H_{\text{dR},c}^j(M) = H^j(\mathcal{A}_c^\bullet(M), d)$, where $\mathcal{A}_c^j(M)$ consists of compactly supported smooth forms. This support condition is also different from square integrability, which will enter when a metric is introduced.

2 A metric on forms and the Hodge star

The de Rham complex needs a smooth structure; integration of top-degree forms additionally needs an orientation. A Riemannian metric supplies lengths, angles, a volume density, and a pointwise inner product on forms; an orientation turns the density into a volume form. None of these constructions requires compactness or completeness. On a compact manifold without boundary, they will ultimately select a unique harmonic representative of each cohomology class.

2.1 Pointwise inner products come first

Let (M^m, g) be a Riemannian manifold. At each point p , the metric is an inner product on the finite-dimensional vector space $T_p M$. The musical isomorphisms are

$$v^\flat = g(v, \cdot), \quad \xi^\sharp = (b)^{-1}\xi,$$

and the induced inner product on $T_p^* M$ is

$$\langle \xi, \eta \rangle_g = g(\xi^\sharp, \eta^\sharp).$$

In coordinates, write $g = \sum g_{ij} dx^i \otimes dx^j$ and let $(g^{ij}) = (g_{ij})^{-1}$. Then

$$\langle dx^i, dx^j \rangle_g = g^{ij}, \quad \left| \sum_i a_i dx^i \right|_g^2 = \sum_{i,j} g^{ij} a_i a_j. \quad (2.1)$$

Thus the metric on covectors uses the *inverse* matrix. For example, if $g = a^2 dx^2 + b^2 dy^2$, then $|dx|_g = a^{-1}$ and $|dy|_g = b^{-1}$; the orthonormal coframe is $(a dx, b dy)$.

On $\Lambda^k T_p^* M$, the induced inner product is characterized by

$$\langle \xi^1 \wedge \cdots \wedge \xi^k, \eta^1 \wedge \cdots \wedge \eta^k \rangle_g = \det(\langle \xi^i, \eta^j \rangle_g)_{i,j=1}^k, \quad (2.2)$$

extended bilinearly. If $(e^1, \dots, e^m)$ is an orthonormal coframe, then the increasing wedge products $e^I = e^{i_1} \wedge \cdots \wedge e^{i_k}$ form an orthonormal basis. Consequently, if $\alpha = \sum_I a_I e^I$ and $\beta = \sum_I b_I e^I$, then

$$\langle \alpha, \beta \rangle_g = \sum_I a_I b_I, \quad |\alpha|_g^2 = \sum_I a_I^2.$$

These are statements in a *single fiber*. For smooth forms, $p \mapsto \langle \alpha_p, \beta_p \rangle_g$ is a smooth function on M , not yet a number obtained by integrating over M .

Suppose also that M is oriented. There is a unique smooth top form dV_g whose value on every positively oriented orthonormal tangent frame is 1. In positive coordinates,

$$dV_g = \sqrt{\det(g_{ij})} dx^1 \wedge \cdots \wedge dx^m; \quad dV_g = e^1 \wedge \cdots \wedge e^m \quad (2.3)$$

in a positive orthonormal coframe. In particular $|dV_g|_g = 1$. Every top form is uniquely $f dV_g$ and has pointwise norm $|f|$. The metric chooses this particular positive volume form; the orientation alone does not.

2.2 Definition and algebra of the star

Definition 2.1 (Hodge star). On an oriented Riemannian m -manifold, the Hodge star is the pointwise linear map $*$: $\Lambda^k T_p^* M \rightarrow \Lambda^{m-k} T_p^* M$ determined by

$$\alpha \wedge * \beta = \langle \alpha, \beta \rangle_g dV_g \quad (\alpha, \beta \in \Lambda^k T_p^* M). \quad (2.4)$$

Applying it at every point gives $*$: $\mathcal{A}^k(M) \rightarrow \mathcal{A}^{m-k}(M)$.

Here is an explicit construction, which proves existence and uniqueness. Choose a positive orthonormal coframe, let I^c be the increasing complement of I , and let $\varepsilon(I)$ be the sign of the permutation (I, I^c) of $(1, \dots, m)$. Then

$$*e^I = \varepsilon(I)e^{I^c}. \quad (2.5)$$

Indeed, $e^J \wedge *e^I = \delta_{IJ} dV_g$. This both verifies the defining identity and determines each coefficient of $*e^I$ uniquely. Uniqueness makes the construction independent of the chosen coframe. The coefficients vary smoothly with g , so $*$ sends smooth forms to smooth forms. It is a *pointwise*, order-zero operator: $*(f\alpha) = f * \alpha$, even when f is nonconstant.

Proposition 2.2 (Basic identities). *For real k -forms α, β ,*

$$*1 = dV_g, \quad *dV_g = 1, \quad (2.6)$$

$$*^2\alpha = (-1)^{k(m-k)}\alpha, \quad \langle *\alpha, *\beta \rangle_g = \langle \alpha, \beta \rangle_g, \quad (2.7)$$

$$\alpha \wedge *\beta = \beta \wedge *\alpha, \quad \alpha \wedge *\alpha = |\alpha|_g^2 dV_g. \quad (2.8)$$

Proof. The complement formula sends an orthonormal basis to an orthonormal basis with possible signs, proving the isometry assertion. Applying it twice exchanges a block of k indices with a block of $m - k$ indices, giving the sign $(-1)^{k(m-k)}$. The remaining statements follow directly from the defining identity and the symmetry of the real inner product. $\square$

Reversing orientation replaces dV_g by $-dV_g$ and therefore replaces $*$ by $-*$. A different transformation is a conformal change of metric: if $\tilde{g} = e^{2u}g$ with the same orientation, then

$$|\alpha|_{\tilde{g}}^2 = e^{-2ku} |\alpha|_g^2, \quad dV_{\tilde{g}} = e^{mu} dV_g, \quad *\tilde{\alpha}|_{\mathcal{A}^k} = e^{(m-2k)u} *_g \alpha. \quad (2.9)$$

The first identity follows from the inverse metric and the determinant formula; substitution in the defining identity proves the third. In middle degree $m = 2k$, the star depends only on the conformal class.

Remark 2.3 (Complex forms and the conjugation convention). Unless otherwise indicated, forms in this topic are real. On complex forms we extend $*$ *complex linearly*, and use the Hermitian inner product linear in its first argument. Thus the defining identity becomes

$$\alpha \wedge *\bar{\beta} = \langle \alpha, \beta \rangle_g dV_g.$$

In an orthonormal coframe the inner product is $\sum_I a_I \bar{b}_I$. Conjugation belongs in the second argument of the pairing, not in the definition of the complex-linear star.

2.3 From pointwise lengths to the global L^2 inner product

On any Riemannian manifold, the following formula defines an inner product on square-integrable smooth forms. With a chosen orientation it can also be written using the star:

$$(\alpha, \beta)_{L^2} = \int_M \langle \alpha, \beta \rangle_g dV_g = \int_M \alpha \wedge *\bar{\beta}, \quad \|\alpha\|_{L^2}^2 = \int_M |\alpha|_g^2 dV_g. \quad (2.10)$$

The conjugation can be omitted for real forms. The pointwise quantity $|\alpha|_g$ measures a covector at p ; the global quantity $\|\alpha\|_{L^2}$ is one number measuring the whole field. If the latter vanishes for a smooth form, continuity forces the form to vanish everywhere, so this is positive definite. Although the oriented formula uses $*$, the inner product is independent of orientation: reversing orientation changes both $*$ and oriented integration by a minus sign. The density expression also defines it on a nonorientable manifold. No boundary condition is needed to define this norm. Compactness enters only when we want *every* smooth form to have finite norm.

On a compact manifold, the space $\mathcal{A}^k(M)$ with this norm is usually *not complete*. For instance, smooth approximations to the characteristic function of an arc on S^1 converge in L^2 to a discontinuous function. Its Hilbert space completion is denoted $L^2\mathcal{A}^k(M)$ and consists of measurable, square-integrable forms, identified when equal almost everywhere. Differentiation on this completion requires a domain and additional regularity; it is not an everywhere-defined continuous linear map on L^2 . On a noncompact manifold the same Hilbert space is obtained by completing $\mathcal{A}_c^k(M)$, the compactly supported smooth forms, or all square-integrable smooth forms. One cannot complete unrestricted $\mathcal{A}^k(M)$ in this norm, since some of its elements have infinite integral. Finite total volume does not by itself make every smooth form square-integrable; an unbounded form may still have divergent norm.

2.4 Flat tori: lengths, energies, and the choice of lattice

Example 2.4 (The rectangular torus, with coordinates of period one). Put $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ and choose the metric $g = A^2 dx^2 + B^2 dy^2$, where $A, B > 0$ are constants. Its two coordinate circles have lengths A, B ; the coordinates still have period 1. The positive orthonormal coframe $(A dx, B dy)$ gives

$$dV_g = AB dx \wedge dy, \quad *dx = \frac{B}{A} dy, \quad *dy = -\frac{A}{B} dx, \quad *(dx \wedge dy) = \frac{1}{AB}. \quad (2.11)$$

For $\alpha = P(x, y)dx + Q(x, y)dy$, with periodic coefficients,

$$|\alpha_{(x,y)}|_g^2 = \frac{P(x, y)^2}{A^2} + \frac{Q(x, y)^2}{B^2}, \quad (2.12)$$

$$\|\alpha\|_{L^2}^2 = \int_0^1 \int_0^1 \left(\frac{B}{A} P(x, y)^2 + \frac{A}{B} Q(x, y)^2 \right) dx dy. \quad (2.13)$$

In particular $\|1\|^2 = AB$, $\|dx\|^2 = B/A$, $\|dy\|^2 = A/B$, and $\|dx \wedge dy\|^2 = 1/(AB)$. These global squared norms include the volume factor; they are not the corresponding pointwise squared norms.

For a nonconstant test take $\eta = \sin(2\pi x) dx = d(-\cos(2\pi x)/(2\pi))$. Then

$$|\eta_{(x,y)}|_g^2 = \frac{\sin^2(2\pi x)}{A^2}, \quad \|\eta\|_{L^2}^2 = \frac{B}{2A}, \quad (dx, \eta)_{L^2} = 0.$$

The pointwise norm vanishes along two circles, while the global norm is positive. On the square torus $A = B = 1$ this gives $\|dx + \eta\|^2 = 3/2$ and $\|dx\|^2 = 1$, even though these forms have the same cohomology class.

This calculation has a general explanation. If $h = c_1 dx + c_2 dy$ has constant coefficients and f is periodic, then

$$(h, df)_{L^2} = \frac{B}{A} c_1 \int_{[0,1]^2} f_x dx dy + \frac{A}{B} c_2 \int_{[0,1]^2} f_y dx dy = 0.$$

Opposite sides of the square cancel each integral. Thus

$$\|h + df\|_{L^2}^2 = \|h\|_{L^2}^2 + \|df\|_{L^2}^2. \quad (2.14)$$

Together with (1.11), this already proves that $c_1 dx + c_2 dy$ is the unique form of least L^2 norm in its class. Uniqueness here is uniqueness of the form: f itself is determined only up to a constant. Once the formal adjoint is defined, this orthogonality to all df will become the equation $d^*h = 0$. It is the first concrete appearance of the condition that defines a harmonic representative.

Example 2.5 (An oblique complex torus). Let $\tau = a + ib$ with $b > 0$, and write

$$T_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}), \quad z = x + \tau y, \quad (x, y) \in \mathbb{R}^2/\mathbb{Z}^2.$$

The Euclidean metric $|dz|^2$ on $\mathbb{C}$ descends to

$$g = (dx + a dy)^2 + b^2 dy^2 = dx^2 + 2a dx dy + (a^2 + b^2) dy^2, \quad dV_g = b dx \wedge dy.$$

Use the positive orthonormal coframe $e^1 = dx + a dy$, $e^2 = b dy$. Since $*e^1 = e^2$ and $*e^2 = -e^1$, solving back for dx, dy gives

$$*dx = \frac{a}{b} dx + \frac{a^2 + b^2}{b} dy, \quad *dy = -\frac{1}{b} dx - \frac{a}{b} dy, \quad *(dx \wedge dy) = \frac{1}{b}. \quad (2.15)$$

The global L^2 Gram matrix in the period-normalized basis dx, dy is

$$\begin{pmatrix} (dx, dx) & (dx, dy) \\ (dy, dx) & (dy, dy) \end{pmatrix}_{L^2} = \frac{1}{b} \begin{pmatrix} a^2 + b^2 & -a \\ -a & 1 \end{pmatrix}. \quad (2.16)$$

For example, $\tau = 1 + i$ gives $*dx = dx + 2dy$, $*dy = -dx - dy$, and $\|c dx + d dy\|^2 = 2c^2 - 2cd + d^2$. The off-diagonal term records that the period-normalized coordinate covectors are not orthogonal. The period pairing is still $\int_{T_\tau} dx \wedge dy = 1$; it is independent of these metric data.

With the complex-linear convention for $*$, we obtain

$$dz = dx + \tau dy = e^1 + ie^2, \quad *dz = -i dz, \quad *d\bar{z} = i d\bar{z}, \quad \|dz\|_{L^2}^2 = 2b. \quad (2.17)$$

The periods of dz on the two coordinate loops are $1, \tau$; dz descends although z does not. All the displayed constant forms are closed, and their stars have constant coefficients and are closed. After defining d^* , we will conclude that they are harmonic; the Fourier argument in Corollary 3.12 will prove that there are no other harmonic forms for a flat torus. The same reasoning applies to the rectangular metric above. Thus these calculations provide explicit candidates for harmonic representatives before invoking the general analytic existence theorem.

2.5 Calculations in the plane and in three dimensions

Example 2.6 (The oriented Euclidean plane). For $g = dx^2 + dy^2$ and orientation $dx \wedge dy$,

$$*1 = dx \wedge dy, \quad *dx = dy, \quad *dy = -dx, \quad *(dx \wedge dy) = 1.$$

Hence $*(a dx + b dy) = -b dx + a dy$ and $*^2 = -1$ on one-forms. For the nonconstant form $\alpha = x dy - y dx$,

$$*\alpha = -x dx - y dy = -\frac{1}{2}d(x^2 + y^2), \quad |\alpha|^2 = x^2 + y^2.$$

On the disk D_R , polar integration gives

$$\|\alpha\|_{L^2(D_R)}^2 = \int_0^{2\pi} \int_0^R r^2 r dr d\phi = \frac{\pi R^4}{2}.$$

The star differentiates no coefficient: the derivative in the displayed expression for $*\alpha$ is simply a convenient way to recognize that form.

Example 2.7 (A coordinate coframe need not be orthonormal). Let $g = E dx^2 + 2F dx dy + G dy^2$, where $E > 0$ and $D = EG - F^2 > 0$, and orient the chart by $dx \wedge dy$. Using the inverse metric in the definition gives

$$*dx = \frac{F dx + G dy}{\sqrt{D}}, \quad *dy = -\frac{E dx + F dy}{\sqrt{D}}, \quad *(dx \wedge dy) = \frac{1}{\sqrt{D}}. \quad (2.18)$$

For example, $g = dx^2 + 2 dx dy + 2 dy^2 = (dx + dy)^2 + dy^2$ has positive orthonormal coframe $(dx + dy, dy)$ and $dV_g = dx \wedge dy$. Here

$$*dx = dx + 2dy, \quad *dy = -dx - dy.$$

For $\beta = x dx + y dy$ this yields

$$*\beta = (x - y)dx + (2x - y)dy, \quad \beta \wedge *\beta = (2x^2 - 2xy + y^2)dx \wedge dy.$$

The coefficient is exactly $|\beta|_g^2$, computed using $g^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$.

Example 2.8 (Euclidean three-space and vector calculus). For $g = dx^2 + dy^2 + dz^2$ and orientation $dx \wedge dy \wedge dz$, all degrees are described by

α	1	dx	dy	dz
$*\alpha$	$dx \wedge dy \wedge dz$	$dy \wedge dz$	$dz \wedge dx$	$dx \wedge dy$

and

α	$dy \wedge dz$	$dz \wedge dx$	$dx \wedge dy$	$dx \wedge dy \wedge dz$
$*\alpha$	dx	dy	dz	1.

Here $*^2 = 1$ in every degree. If $X = P\partial_x + Q\partial_y + R\partial_z$, then

$$\begin{aligned} *X^\flat &= P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy, \\ *dX^\flat &= (R_y - Q_z)dx + (P_z - R_x)dy + (Q_x - P_y)dz, \\ d(*X^\flat) &= (P_x + Q_y + R_z) dx \wedge dy \wedge dz. \end{aligned}$$

Thus $(\text{curl } X)^\flat = *dX^\flat$ and $\text{div } X = *d * X^\flat$. Also $(X \times Y)^\flat = *(X^\flat \wedge Y^\flat)$, which makes the dependence of the cross product on both the metric and orientation explicit.

Example 2.9 (The same calculations on a three-torus). On $T^3 = \mathbb{R}^3/\mathbb{Z}^3$ with its standard flat metric, the constant forms dx, dy, dz and all their wedge products descend, and the entire preceding star table is a table of global forms. The volume of this torus is 1, so these forms are orthonormal for the global L^2 inner product as well as pointwise. For a vector field with periodic coefficients, the curl and divergence formulas above descend too. Periodicity makes the fluxes across opposite faces cancel, for example

$$\int_{[0,1]^3} (P_x + Q_y + R_z) dx dy dz = 0.$$

The constant forms and their stars are closed. Once the analytic argument in Theorem 3.11 is established, these will be exactly the harmonic forms, giving harmonic-space dimensions $(1, 3, 3, 1)$.

2.6 Four dimensions: self-dual and anti-self-dual two-forms

In oriented Euclidean $\mathbb{R}^4$, write $e^i = dx^i$, $e^{ij} = e^i \wedge e^j$, and $dV = e^{1234}$. The complete table in degree two is

α	e^{12}	e^{13}	e^{14}	e^{23}	e^{24}	e^{34}
$*\alpha$	e^{34}	$-e^{24}$	e^{23}	e^{14}	$-e^{13}$	e^{12} .

For example the minus sign in $*e^{13} = -e^{24}$ compensates for the single transposition in $e^{13} \wedge e^{24} = -dV$. Since $*^2 = 1$ on two-forms, there is a decomposition

$$\Lambda^2 = \Lambda_+^2 \oplus \Lambda_-^2, \quad \Lambda_\pm^2 = \{\alpha : *\alpha = \pm\alpha\}, \quad \alpha_\pm = \frac{1}{2}(\alpha \pm *\alpha). \quad (2.19)$$

Explicit orthonormal bases are

$$\begin{aligned}\Lambda_+^2 &: \frac{e^{12} + e^{34}}{\sqrt{2}}, \quad \frac{e^{13} - e^{24}}{\sqrt{2}}, \quad \frac{e^{14} + e^{23}}{\sqrt{2}}, \\ \Lambda_-^2 &: \frac{e^{12} - e^{34}}{\sqrt{2}}, \quad \frac{e^{13} + e^{24}}{\sqrt{2}}, \quad \frac{e^{14} - e^{23}}{\sqrt{2}}.\end{aligned}$$

Both spaces have dimension 3. The star is an isometric involution, so on two-forms it is self-adjoint:

$$\langle *\alpha, \beta \rangle = \langle \alpha, *\beta \rangle.$$

Its two eigenspaces are orthogonal. In particular $\alpha_+ \wedge \alpha_- = 0$, and

$$\alpha \wedge \alpha = (|\alpha_+|^2 - |\alpha_-|^2)dV. \quad (2.20)$$

More generally the symmetric wedge pairing B , defined by $\alpha \wedge \beta = B(\alpha, \beta)dV$, satisfies $B(\alpha, \beta) = \langle \alpha, *\beta \rangle$ and has signature $(3, 3)$. For $\alpha = e^{12} + 2e^{34}$,

$$\alpha_+ = \frac{3}{2}(e^{12} + e^{34}), \quad \alpha_- = -\frac{1}{2}(e^{12} - e^{34}), \quad \alpha \wedge \alpha = 4dV,$$

in agreement with $|\alpha_+|^2 - |\alpha_-|^2 = 9/2 - 1/2 = 4$.

Remark 2.10 (Two different meanings of symmetry). Every two-form is an *alternating* bilinear tensor: $\alpha(v, w) = -\alpha(w, v)$. The terms *self-dual* and *anti-self-dual* describe its eigenvalue under $*$; they do not mean that the tensor is symmetric or antisymmetric under exchange of v, w . The symmetry above belongs to the operator $*$ and to the wedge pairing of two-forms, since $\alpha \wedge \beta = \beta \wedge \alpha$ in degree two. On a compact oriented four-manifold without boundary the corresponding splitting of harmonic two-forms will give the positive and negative spaces of the cohomological intersection pairing. Their dimensions need not each be 3: the number 3 here is the dimension in one cotangent fiber.

All these constructions work pointwise on any oriented Riemannian four-manifold. The bundles $\Lambda_\pm^2 T^*M$ have rank 3 and are unchanged by conformal rescaling, by (2.9). Reversing orientation exchanges them.

Example 2.11 (Global self-duality on the flat four-torus). On $T^4 = \mathbb{R}^4/\mathbb{Z}^4$ with its standard metric and orientation, $e^i = dx^i$ are global, although the real coordinates x^i are not. The six constant two-forms displayed above are therefore global self-dual and anti-self-dual forms. Their coefficients and their stars are constant, so they will be harmonic. The Fourier calculation in Theorem 3.11 will show that all harmonic two-forms are constant. Consequently

$$\mathcal{H}^2(T^4) = \text{span}_{\mathbb{R}}\{e^{12}, e^{13}, e^{14}, e^{23}, e^{24}, e^{34}\}, \quad b_2^+ = b_2^- = 3.$$

Here $b_2^\pm$ denote the dimensions of the ± 1 eigenspaces of $*$ among harmonic forms. Since $\int_{T^4} dV = 1$, the displayed normalized self-dual and anti-self-dual bases have intersection matrices I_3 and $-I_3$, respectively, and are mutually orthogonal.

Pointwise self-duality alone imposes no differential equation. For example, the global form

$$\alpha = \sin(2\pi x^1)(e^{12} + e^{34})$$

satisfies $*\alpha = \alpha$, but $d\alpha = 2\pi \cos(2\pi x^1)e^{134} \neq 0$, so it is not harmonic. The bundle of self-dual two-forms has rank 3, its smooth section space is infinite-dimensional, and its harmonic subspace on this particular torus has dimension 3. These are three different statements.

2.7 The round sphere: local calculations and global forms

On the unit sphere $S^2 \subset \mathbb{R}^3$, use the outward orientation and spherical coordinates

$$(x, y, z) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).$$

Away from the poles, the round metric and positive orthonormal coframe are

$$g = d\theta^2 + \sin^2 \theta d\phi^2, \quad e^1 = d\theta, \quad e^2 = \sin \theta d\phi.$$

Hence

$$dV_g = \sin \theta d\theta \wedge d\phi, \quad *d\theta = \sin \theta d\phi, \quad *d\phi = -\frac{d\theta}{\sin \theta}, \quad *(d\theta \wedge d\phi) = \frac{1}{\sin \theta}. \quad (2.21)$$

In particular

$$|a d\theta + b d\phi|_g^2 = a^2 + \frac{b^2}{\sin^2 \theta}.$$

These are coordinate formulas on appropriate charts, not global smooth one-forms $d\theta, d\phi$ at the poles. The global volume form is

$$dV_g = (x dy \wedge dz - y dx \wedge dz + z dx \wedge dy)|_{S^2},$$

the contraction of ambient Euclidean volume with the unit outward normal. It satisfies $*dV_g = 1$, and $\|dV_g\|_{L^2}^2 = \text{Area}(S^2) = 4\pi$.

For a smooth global one-form, take $dz = d(z|_{S^2})$. Using $z = \cos \theta$ gives

$$\begin{aligned} dz &= -\sin \theta d\theta, & *dz &= -\sin^2 \theta d\phi = (y dx - x dy)|_{S^2}, \\ |dz|_g^2 &= \sin^2 \theta = 1 - z^2, & \|dz\|_{L^2}^2 &= 2\pi \int_0^\pi \sin^3 \theta d\theta = \frac{8\pi}{3}. \end{aligned}$$

The ambient expression verifies smoothness of $*dz$ even at the poles. The pointwise norm vanishes there because z has a maximum and a minimum there; this is a geometric fact, not a coordinate defect.

2.8 The hyperbolic upper half-plane and quotients

Let $\mathbb{H} = \{x + iy : y > 0\}$, with its usual orientation and hyperbolic metric

$$g_{\text{hyp}} = \frac{dx^2 + dy^2}{y^2}.$$

The positive orthonormal coframe is $e^1 = dx/y$, $e^2 = dy/y$. Therefore

$$dV_{\text{hyp}} = y^{-2} dx \wedge dy, \quad |a dx + b dy|_{\text{hyp}}^2 = y^2(a^2 + b^2), \quad (2.22)$$

$$*1 = y^{-2} dx \wedge dy, \quad *dx = dy, \quad *dy = -dx, \quad (2.23)$$

$$*(f dx \wedge dy) = f y^2. \quad (2.24)$$

The star on one-forms agrees with the Euclidean star, although their pointwise norms do not agree. For a real one-form, the factors cancel in the energy density:

$$|a dx + b dy|_{\text{hyp}}^2 dV_{\text{hyp}} = (a^2 + b^2) dx dy. \quad (2.25)$$

This is the conformal invariance of the L^2 norm on one-forms in dimension two. For comparison, functions have density $f^2 y^{-2} dx dy$, and two-forms $f dx \wedge dy$ have density $f^2 y^2 dx dy$.

If $\Gamma \subset \mathrm{PSL}_2(\mathbb{R})$ is a discrete group acting freely on $\mathbb{H}$, then $\Sigma = \Gamma \backslash \mathbb{H}$ is an oriented hyperbolic surface. A form upstairs descends precisely when it is Γ -invariant. Orientation-preserving isometries commute with $*$, so the metric, volume, and star descend. The coordinate forms dx and dy generally do *not* descend for an arbitrary Γ . For a descended one-form, its global L^2 norm can be computed by integrating (2.25) over a fundamental domain.

For a concrete calculation take the translation quotient $\langle z \mapsto z + 1 \rangle \backslash \mathbb{H}$. Both dx and dy descend, and over the strip $0 \leq x \leq 1$, $a \leq y \leq b$,

$$\mathrm{Area}_{\mathrm{hyp}} = \frac{1}{a} - \frac{1}{b}, \quad \|dx\|_{L^2}^2 = b - a.$$

The cusp region $y \geq a$ has finite area $1/a$, while dx has infinite L^2 norm on it. Thus finite volume alone does not ensure that a smooth form is square-integrable. When we apply the compact Hodge theorem below to a hyperbolic quotient, we assume in addition that Γ is *cocompact*, so that Σ is compact without boundary.

3 The adjoint, the Laplacian, and the analytic bridge

Let M be an oriented, smooth Riemannian n -manifold without boundary, not necessarily compact. Forms are real unless otherwise indicated. The codifferential, Laplacian, and their differential identities make sense on every such M . Global integration and existence statements will specify compactness or support conditions. We write $(\alpha, \beta)_{L^2} = \int_M \alpha \wedge * \beta$ when the integral exists and use the *nonnegative* convention for the Laplacian.

3.1 Deriving the codifferential from Stokes' theorem

Definition 3.1 (Formal adjoint). The codifferential $d^* : \mathcal{A}^k(M) \rightarrow \mathcal{A}^{k-1}(M)$ is the differential operator characterized by

$$(d\alpha, \beta)_{L^2} = (\alpha, d^*\beta)_{L^2}, \quad \alpha \in \mathcal{A}_c^{k-1}(M), \quad \beta \in \mathcal{A}^k(M).$$

Here $\mathcal{A}_c^{k-1}(M)$ denotes compactly supported smooth forms, so both integrals exist. If M is compact, every smooth form is compactly supported and this restriction is automatic. On functions, $d^* = 0$. This is a *formal* adjoint, defined by testing smooth forms; an adjoint of an unbounded operator on L^2 also requires specifying its operator domain.

Proposition 3.2 (The sign of d^*). *On k -forms on any oriented Riemannian manifold,*

$$d^* = (-1)^{n(k+1)+1} * d *. \quad (3.1)$$

In particular, $(d^)^2 = 0$.*

Proof. For compactly supported α of degree $k-1$, Stokes gives

$$0 = \int_M d(\alpha \wedge * \beta) = \int_M d\alpha \wedge * \beta + (-1)^{k-1} \int_M \alpha \wedge d(*\beta).$$

Thus we need $*(d^*\beta) = (-1)^k d(*\beta)$. The form $d(*\beta)$ has degree $n-k+1$, so the inverse of $*$ in this equation contributes the sign $(-1)^{(k-1)(n-k+1)}$. Therefore

$$d^*\beta = (-1)^{k+(k-1)(n-k+1)} * d * \beta = (-1)^{n(k+1)+1} * d * \beta.$$

The exponents differ by an even integer. This constructs the formal adjoint; uniqueness follows by testing against arbitrary compactly supported forms and using the positive definite pointwise pairing. Composing the displayed formula in consecutive degrees reduces $(d^*)^2$ to a sign times $*d^2*$, hence to zero. $\square$

Example 3.3 (Divergence and a coordinate sign check). On an oriented Euclidean plane, $*dx = dy$, $*dy = -dx$, and $*(dx \wedge dy) = 1$. For $\alpha = P dx + Q dy$ and $\beta = R dx \wedge dy$,

$$\begin{aligned} d\alpha &= (Q_x - P_y) dx \wedge dy, & d^*\alpha &= -(P_x + Q_y), \\ d^*\beta &= R_y dx - R_x dy. \end{aligned}$$

For instance, $d^*(x^2 dx + xy dy) = -3x$. Adding $dd^*\alpha$ and $d^*d\alpha$ cancels the mixed derivatives and gives

$$\Delta\alpha = -(P_{xx} + P_{yy}) dx - (Q_{xx} + Q_{yy}) dy.$$

In any dimension, $d^*\alpha = -\operatorname{div}(\alpha^\sharp)$ for a one-form. These formulas are local: on a noncompact space, the adjoint identity is first justified using compactly supported test forms.

Example 3.4 (The same operators on the flat torus). Let $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ carry $g_0 = dx^2 + dy^2$ and the orientation $dx \wedge dy$. A form on T^2 is a form on $\mathbb{R}^2$ whose coefficients are periodic with period one in both variables. Thus the preceding formulas apply to periodic P, Q, R . In particular,

$$\Delta(P dx + Q dy) = (-P_{xx} - P_{yy}) dx + (-Q_{xx} - Q_{yy}) dy.$$

For $\alpha = \sin(2\pi x) dx + \cos(2\pi x) dy$,

$$\begin{aligned} d\alpha &= -2\pi \sin(2\pi x) dx \wedge dy, & d^*\alpha &= -2\pi \cos(2\pi x), \\ dd^*\alpha &= 4\pi^2 \sin(2\pi x) dx, & d^*d\alpha &= 4\pi^2 \cos(2\pi x) dy. \end{aligned}$$

Hence $\Delta\alpha = 4\pi^2\alpha$. The two summands already separate an exact form from a coexact form:

$$\alpha = d\left(-\frac{\cos(2\pi x)}{2\pi}\right) + d^*\left(-\frac{\sin(2\pi x)}{2\pi} dx \wedge dy\right).$$

Opposite edges of the unit square cancel in integration by parts; there is no boundary term on the quotient. The expression $x^2 dx$ from the plane example does not descend to this torus, whereas every form in this example does.

3.2 The Laplacian and what follows without analysis

Definition 3.5. The Hodge Laplacian and the space of harmonic forms are

$$\Delta_k = dd^* + d^*d : \mathcal{A}^k(M) \longrightarrow \mathcal{A}^k(M), \quad \mathcal{H}^k(M, g) = \ker \Delta_k.$$

Subscripts are suppressed when the degree is clear. This definition does not require compactness: $\mathcal{H}^k(M, g)$ denotes the smooth kernel, with no L^2 condition imposed, and it need not be finite dimensional.

If M is compact without boundary, integration by parts gives, for arbitrary smooth α, β of the same degree,

$$(\Delta\alpha, \beta)_{L^2} = (d\alpha, d\beta)_{L^2} + (d^*\alpha, d^*\beta)_{L^2} = (\alpha, \Delta\beta)_{L^2}. \quad (3.2)$$

The same identity holds on arbitrary M if at least one of the two forms is compactly supported. In particular, for every smooth α on compact M , or every compactly supported smooth α on arbitrary M ,

$$(\Delta\alpha, \alpha)_{L^2} = \|d\alpha\|_{L^2}^2 + \|d^*\alpha\|_{L^2}^2, \quad \Delta\alpha = 0 \iff d\alpha = d^*\alpha = 0. \quad (3.3)$$

The implication from left to right uses a global integral and positivity; it is not a pointwise separation of the two terms of Δ . The reverse implication, $d\alpha = d^*\alpha = 0 \Rightarrow \Delta\alpha = 0$, holds on every M , without support or integrability assumptions. For a function,

$$\Delta f = d^*df = -\operatorname{div}(\nabla f), \quad \Delta_{\mathbb{R}^n} f = -\sum_j \partial_j^2 f.$$

Consequently, harmonic functions on a connected compact manifold without boundary are constant: (3.3) forces $df = 0$. On $\mathbb{R}$, however, $f(x) = x$ satisfies $\Delta f = -f'' = 0$ but $df = dx \neq 0$. Thus unrestricted smooth harmonic forms on a noncompact manifold need not be closed. On the compact interval $[0, 1]$, the same function shows why a boundary cannot be ignored: integration by parts has a boundary term unless suitable conditions are imposed.

Example 3.6 (Finding all harmonic forms on the square torus). For a periodic one-form $P dx + Q dy$, integration by parts gives

$$(\Delta\alpha, \alpha)_{L^2} = \int_{[0,1]^2} (P_x^2 + P_y^2 + Q_x^2 + Q_y^2) dx dy.$$

If $\Delta\alpha = 0$, all four derivatives vanish, so P, Q are constant. Conversely, constant coefficients give $d\alpha = d^*\alpha = 0$. The same argument applies to functions and to the coefficient of a two-form. Thus, for this particular metric,

$$\mathcal{H}^0(T^2, g_0) = \mathbb{R} \cdot 1, \quad \mathcal{H}^1(T^2, g_0) = \text{span}_{\mathbb{R}}\{dx, dy\}, \quad \mathcal{H}^2(T^2, g_0) = \mathbb{R} \cdot (dx \wedge dy).$$

This computation uses periodic integration and positivity, with no appeal to the general Hodge theorem. Below we will also prove directly that these spaces give every de Rham cohomology class.

Proposition 3.7 (Commutations and orthogonality). *On smooth forms on any oriented Riemannian manifold,*

$$d\Delta = \Delta d, \quad d^*\Delta = \Delta d^*, \quad *\Delta = \Delta*.$$

If M is compact without boundary, the spaces $\mathcal{H}^k$, $d\mathcal{A}^{k-1}$, and $d^\mathcal{A}^{k+1}$ are pairwise L^2 orthogonal.*

Proof. The first two identities follow by expansion and $d^2 = (d^*)^2 = 0$; for example $d\Delta = dd^*d = \Delta d$. Formula (3.1) and the square of $*$ give, for a k -form α ,

$$d(*\alpha) = (-1)^k * (d^*\alpha), \quad d^*(*\alpha) = (-1)^{k+1} * (d\alpha).$$

Applying these in consecutive degrees yields $*dd^*\alpha = d^*d(*\alpha)$ and $*d^*d\alpha = dd^*(*\alpha)$; add them. For orthogonality, now assume M is compact without boundary. Then $h \in \mathcal{H}^k$ satisfies $dh = d^*h = 0$ by (3.3), so

$$(h, da) = (d^*h, a) = 0, \quad (h, d^*b) = (dh, b) = 0, \quad (da, d^*b) = (d^2a, b) = 0.$$

□

On a compact manifold without boundary, what remains is *existence*: do these three orthogonal spaces span every smooth form? The next finite dimensional argument isolates the linear algebra that we want to recover.

3.3 A complete finite dimensional Hodge theorem

Let $(C^\bullet, d)$ be a complex of finite dimensional real inner product spaces. Write $d_k : C^k \rightarrow C^{k+1}$ and let d_k^* be its ordinary adjoint matrix. Set

$$\Delta_k = d_{k-1}d_{k-1}^* + d_k^*d_k, \quad \mathcal{K}^k = \ker \Delta_k.$$

Theorem 3.8 (Finite dimensional model). *There is an orthogonal decomposition*

$$C^k = \text{im } d_{k-1} \oplus \mathcal{K}^k \oplus \text{im } d_k^*.$$

Moreover, $\mathcal{K}^k = \ker d_k \cap \ker d_{k-1}^$ and $h \mapsto [h]$ induces an isomorphism $\mathcal{K}^k \simeq \ker d_k / \text{im } d_{k-1}$.*

Proof. For every $v \in C^k$,

$$(\Delta_k v, v) = \|d_{k-1}^* v\|^2 + \|d_k v\|^2.$$

Thus $\ker \Delta_k = \ker d_{k-1}^* \cap \ker d_k$. Put $E = \operatorname{im} d_{k-1}$ and $F = \operatorname{im} d_k^*$. If $a \in C^{k-1}$ and $b \in C^{k+1}$, then

$$(d_{k-1} a, d_k^* b) = (d_k d_{k-1} a, b) = 0,$$

so $E \perp F$. A vector v is orthogonal to $E + F$ precisely when

$$(v, d_{k-1} a) = 0 \text{ for all } a, \quad (v, d_k^* b) = 0 \text{ for all } b.$$

By the adjoint identities, these conditions are $d_{k-1}^* v = 0$ and $d_k v = 0$, respectively. Hence $(E + F)^\perp = \mathcal{K}^k$. A finite dimensional inner product space is the sum of a subspace and its orthogonal complement, proving the decomposition. Its components are unique because they are orthogonal.

If $d_k v = 0$, decompose $v = e + h + f$ with $f = d_k^* b$. Orthogonality and the adjoint identity give

$$\|f\|^2 = (v, f) = (d_k v, b) = 0.$$

Thus $v = e + h$, proving that every cohomology class has a harmonic representative. If $h = d_{k-1} a$ is harmonic, then $\|h\|^2 = (d_{k-1}^* h, a) = 0$, proving uniqueness and injectivity. $\square$

Example 3.9 (A matrix complex with three nonzero components). Give each space its standard Euclidean inner product and take

$$0 \longrightarrow \mathbb{R} \xrightarrow{A} \mathbb{R}^3 \xrightarrow{B} \mathbb{R} \longrightarrow 0, \quad A = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & 2 \end{pmatrix}.$$

Since $BA = 0$, this is a complex. In degree one,

$$\Delta_1 = AA^\top + B^\top B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

Its orthogonal eigendirections are $(1, 1, 0)$, $(1, -1, 0)$, and $(0, 0, 1)$, with eigenvalues 2, 0, 4. In full detail,

$$(a, b, c) = \underbrace{\left(\frac{a+b}{2}, \frac{a+b}{2}, 0 \right)}_{\operatorname{im} A} + \underbrace{\left(\frac{a-b}{2}, \frac{b-a}{2}, 0 \right)}_{\ker \Delta_1} + \underbrace{(0, 0, c)}_{\operatorname{im} B^\top}.$$

For $(a, b, c) = (3, 1, 2)$, these are $(2, 2, 0)$, $(1, -1, 0)$, and $(0, 0, 2)$, all nonzero. A vector is a cocycle exactly when $c = 0$; its class is then determined by its $(1, -1, 0)$ component.

3.4 Why infinite dimension requires a new theorem

For this comparison, let M be compact without boundary and of positive dimension. The smooth space $\mathcal{A}^k(M)$, for $0 \leq k \leq n$, is not complete in its L^2 norm. Its completion is the Hilbert space $L^2 \mathcal{A}^k(M)$, whose elements need not be smooth. On a noncompact manifold, not every smooth form is in L^2 ; one can instead construct $L^2 \mathcal{A}^k(M)$ by completing $\mathcal{A}_c^k(M)$. Even in a complete Hilbert space, an operator image need not be closed. Orthogonal complements recover $\overline{\operatorname{im} T}$ rather than necessarily $\operatorname{im} T$.

Example 3.10 (A dense image that is not closed). On the Hilbert space ℓ^2 , define

$$T(x_1, x_2, \dots) = (x_1, x_2/2, x_3/3, \dots).$$

This bounded self-adjoint operator is injective. Its image contains every finite sequence and is therefore dense. However, $y = (1, 1/2, 1/3, \dots)$ belongs to ℓ^2 and is not in $\text{im } T$: the only possible preimage is $(1, 1, 1, \dots) \notin \ell^2$. The finite truncations of y lie in $\text{im } T$ and converge to y . Thus $(\text{im } T)^\perp = \ker T = 0$ does not imply $\text{im } T = \ell^2$. In the two-term complex $\ell^2 \xrightarrow{T} \ell^2$, the degree-one Laplacian is TT^* and has zero kernel, although the algebraic cohomology $\ell^2/\text{im } T$ is nonzero. This is the precise gap in an unqualified infinite dimensional version of the preceding proof.

3.5 The flat torus: Fourier analysis makes the bridge explicit

Before introducing an analytic black box, we prove the needed facts on $T^n = \mathbb{R}^n/\mathbb{Z}^n$, equipped with $g_0 = \sum_j (dx^j)^2$ and volume one. The point of this model is to separate two questions: what happens at each frequency, and whether the resulting infinite series defines a smooth solution. The first is finite dimensional linear algebra; the second is analysis.

For $m = (m_1, \dots, m_n) \in \mathbb{Z}^n$, put

$$e_m(x) = e^{2\pi i m \cdot x}, \quad m^\flat = \sum_j m_j dx^j, \quad \iota_m = \iota_{\sum_j m_j \partial_{x^j}}, \quad \lambda_m = 4\pi^2 |m|^2.$$

We temporarily complexify forms and use the Hermitian extension of the inner products, so $(\alpha, \beta)_{L^2} = \int \alpha \wedge \bar{*}\beta$. Write a k -form as

$$\alpha = \sum_{|I|=k} \alpha_I(x) dx^I = \sum_{m \in \mathbb{Z}^n} e_m(x) a_m, \quad a_m = \sum_{|I|=k} \hat{\alpha}_I(m) dx^I,$$

where I is increasing and $\hat{\alpha}_I(m) = \int_{[0,1]^n} \alpha_I(x) e^{-2\pi i m \cdot x} dx$. For real forms, $a_{-m} = \overline{a_m}$; all operators below preserve this condition, so the final answers are real.

The basic Fourier facts and why smoothness matters. The functions e_m form a complete orthonormal basis of $L^2(T^n)$. Completeness follows, for example, by uniformly approximating continuous periodic functions by trigonometric polynomials and using their density in L^2 . Parseval's identity, applied coefficient by coefficient, gives

$$\|\alpha\|_{L^2}^2 = \sum_m |a_m|^2. \quad (3.4)$$

A form is smooth exactly when its Fourier coefficients are *rapidly decreasing*: for every $N \geq 0$ there is C_N with

$$|a_m| \leq C_N (1 + |m|)^{-N} \quad (m \in \mathbb{Z}^n).$$

For the forward direction, integrate by parts repeatedly, or apply $(1 - \sum_j \partial_j^2)^N$ to the smooth coefficient and take its Fourier coefficient. This multiplies a_m by $(1 + \lambda_m)^N$ and leaves a bounded Fourier coefficient. Conversely, a derivative of order r multiplies a_m by at most a constant times $|m|^r$. Choosing $N > n + r$ makes the differentiated series absolutely and uniformly convergent. Thus every derivative exists and can be computed term by term. This criterion will justify each Fourier construction, including the primitives used to compute cohomology.

Each nonzero frequency is a finite dimensional complex. The vector space $e_m \Lambda^\bullet(\mathbb{C}^n)^*$ is preserved by d , d^* , and Δ . On it,

$$\begin{aligned} d(e_m a) &= 2\pi i e_m (m^\flat \wedge a), \\ d^*(e_m a) &= -2\pi i e_m \iota_m a, \\ \Delta(e_m a) &= \lambda_m e_m a. \end{aligned} \tag{3.5}$$

The first formula differentiates e_m ; the second uses the Euclidean adjoint $d^* = -\sum_j \iota_{\partial_{x_j}} \partial_{x_j}$. The third follows from the elementary identity

$$m^\flat \wedge \iota_m a + \iota_m (m^\flat \wedge a) = |m|^2 a. \tag{3.6}$$

This is precisely the finite dimensional model above, now with differential matrix $2\pi i m^\flat \wedge$. For $m \neq 0$ its Laplacian is a positive scalar matrix, and its harmonic subspace is zero. More explicitly, the degree -1 operator

$$K_m = -\frac{i}{2\pi|m|^2} \iota_m$$

satisfies $d_m K_m + K_m d_m = \text{id}$, where $d_m = 2\pi i m^\flat \wedge$. Thus every closed vector in a nonzero mode is exact. At $m = 0$, both differentials vanish and every constant wedge is harmonic.

Theorem 3.11 (Harmonic forms and cohomology of the flat torus). *For the standard flat metric on T^n , in every degree $0 \leq k \leq n$,*

$$\mathcal{H}^k(T^n, g_0) = \text{span}_{\mathbb{R}}\{dx^{i_1} \wedge \cdots \wedge dx^{i_k} : i_1 < \cdots < i_k\}, \quad \dim \mathcal{H}^k(T^n, g_0) = \binom{n}{k}.$$

Every smooth closed form has a unique harmonic representative, namely its coefficientwise average

$$\Pi \alpha = a_0 = \sum_{|I|=k} \left(\int_{[0,1]^n} \alpha_I(x) dx \right) dx^I.$$

Consequently $\mathcal{H}^k(T^n, g_0) \rightarrow H_{\text{dR}}^k(T^n)$ is an isomorphism. This statement is proved independently of the general Hodge theorem.

Proof. Equation (3.5) says that $\Delta \alpha = 0$ if and only if $\lambda_m a_m = 0$ for every m . Thus precisely the zero mode survives. If $d\alpha = 0$, each coefficient satisfies $d_m a_m = 0$. The homotopy identity then gives

$$\alpha - a_0 = d\beta, \quad \beta = \sum_{m \neq 0} e_m K_m a_m.$$

Since $\|K_m\| \leq (2\pi|m|)^{-1}$, the coefficients of β remain rapidly decreasing. The series therefore defines a smooth form and may be differentiated term by term. This proves existence, including the analytic justification for summing the primitives. For $k = 0$ the contraction is zero, and the same identity simply says that a closed function has no nonzero modes.

Finally, an exact form has zero constant Fourier coefficient: every derivative of a periodic function has integral zero. Hence a constant form that is exact must vanish. This proves uniqueness and injectivity; the displayed constant wedges give the dimension. $\square$

Corollary 3.12 (Any constant flat metric on the torus). *If $g_A = \sum_{i,j} A_{ij} dx^i dx^j$ for a constant symmetric positive definite matrix A , the harmonic forms on T^n are still precisely the constant forms. The harmonic representative of a closed form is still its coefficientwise average.*

Proof. The volume density and the inner products on constant wedges are constant. Integration by parts gives

$$d_{g_A}^*(e_m a) = -2\pi i e_m \iota_{A^{-1}m} a, \quad \Delta_{g_A}(e_m a) = 4\pi^2 (m^\top A^{-1} m) e_m a.$$

The quadratic form $m^\top A^{-1} m$ is positive for $m \neq 0$ and comparable to $|m|^2$. Replacing $\iota_m/|m|^2$ by $\iota_{A^{-1}m}/(m^\top A^{-1} m)$ in the preceding proof therefore preserves both the homotopy identity and the rapid-decay argument. Different Fourier modes remain orthogonal, so projection onto constant forms is again coefficientwise averaging. $\square$

Returning to the standard metric g_0 , an arbitrary smooth form has the following explicit orthogonal splitting of each $m \neq 0$ coefficient:

$$a_m = \underbrace{\frac{m^b \wedge \iota_m a_m}{|m|^2}}_{\text{exact mode}} + \underbrace{\frac{\iota_m(m^b \wedge a_m)}{|m|^2}}_{\text{coexact mode}}. \quad (3.7)$$

Wedge multiplication and contraction are adjoints, and $d_m^2 = 0$, so these summands are orthogonal. Both coefficient operators have norm at most one; their sums are therefore smooth. Together with a_0 , they already prove Hodge decomposition on the flat torus. The general theory must supply a replacement for this explicit control of all modes.

3.6 Poisson's equation on the torus: obstruction, inverse, estimates

First find the obstruction, then construct the solution. For $\Delta u = f$ on T^n , comparison of coefficients forces

$$\lambda_m \hat{u}(m) = \hat{f}(m).$$

The $m = 0$ equation is $0 = \hat{f}(0)$. For functions this is the necessary condition $\int_{T^n} f dx = 0$, also obtained by integrating $\Delta u = -\sum_j \partial_j^2 u$. In particular, $\Delta u = 1$ has no periodic solution, even in the distributional sense. For forms, every constant coefficient of f must vanish, equivalently $f \perp \mathcal{H}^k(T^n, g_0)$. If this condition holds, the only possible solution with zero average is

$$u = G_0 f = \sum_{m \neq 0} \frac{\hat{f}(m)}{4\pi^2 |m|^2} e_m. \quad (3.8)$$

Here the coefficients may be scalars or constant wedges. For arbitrary f , the same formula defines $G_0 f$ and satisfies $\Delta G_0 f = f - \Pi f$. All other solutions, when they exist, differ by a constant form. If f is smooth, division by $|m|^2$ preserves rapid decay, so (3.8) is a smooth solution.

Example 3.13 (A periodic Poisson equation with two frequencies). On T^2 , let $f(x, y) = \sin(2\pi x) + \cos(4\pi y)$, which has integral zero. Since

$$\Delta \sin(2\pi x) = 4\pi^2 \sin(2\pi x), \quad \Delta \cos(4\pi y) = 16\pi^2 \cos(4\pi y),$$

the unique solution of zero average is

$$u(x, y) = \frac{\sin(2\pi x)}{4\pi^2} + \frac{\cos(4\pi y)}{16\pi^2}.$$

The second term oscillates twice as fast and its inverse multiplier is four times smaller. This is the concrete meaning of gaining two derivatives. For $f dx$, the solution is $u dx$, because the flat Laplacian acts separately on each coefficient.

The operator domain and the gain of derivatives. For real s , define the Fourier Sobolev norm by

$$\|\alpha\|_{H^s}^2 = \sum_m (1 + \lambda_m)^s |a_m|^2.$$

For nonnegative integers s it is equivalent to the norm defined using weak derivatives up to order s . The completion of smooth forms in this norm is $H^s \mathcal{A}^k(T^n)$. For $s = 0$ this is L^2 by Parseval. The multiplier formula immediately gives

$$\|\Delta u\|_{H^s} \leq \|u\|_{H^{s+2}}.$$

Thus $\Delta : H^{s+2} \rightarrow H^s$ is bounded. Viewed as an operator on L^2 with domain H^2 , it is unbounded: the norm-one functions $e_{(j,0,\dots,0)}$ have Laplacian norm $4\pi^2 j^2 \rightarrow \infty$. One cannot treat it as a bounded matrix on the entire L^2 space.

The first positive eigenvalue is $\lambda_1 = 4\pi^2$, because a nonzero integral frequency has $|m| \geq 1$. For any $f \in H^s$ the inverse formula gives the quantitative estimate

$$\|G_0 f\|_{H^{s+2}} \leq \left(1 + \frac{1}{4\pi^2}\right) \|f - \Pi f\|_{H^s}. \quad (3.9)$$

Indeed, $(1 + \lambda_m)/\lambda_m \leq 1 + (4\pi^2)^{-1}$ for $m \neq 0$; square this inequality and sum the weighted coefficients. This proves existence in H^2 for every L^2 right-hand side orthogonal to the constants. It also proves regularity: if a distributional solution has $\Delta u \in H^s$, its nonzero coefficients lie in H^{s+2} by the same estimate, and its constant coefficient is already smooth. If the right-hand side is smooth, this holds in every Sobolev degree, or directly by rapid decay, giving a smooth solution.

The spectral gap prevents almost harmonic nonconstant forms. For a smooth form with $\Pi\alpha = 0$, Parseval and the energy identity give the sharp Poincaré inequality

$$4\pi^2 \|\alpha\|_{L^2}^2 \leq \sum_{m \neq 0} \lambda_m |a_m|^2 = \|d\alpha\|_{L^2}^2 + \|d^* \alpha\|_{L^2}^2. \quad (3.10)$$

For a function, the right side is $\int |\nabla f|^2$. Equality holds for any form supported on frequencies with $|m| = 1$, for example $\sin(2\pi x^1) dx^I$. A second immediate estimate is

$$\|G_0 f\|_{L^2} \leq \frac{1}{4\pi^2} \|f - \Pi f\|_{L^2}.$$

The gap therefore bounds the inverse and rules out a norm-one sequence orthogonal to constants whose Laplacians tend to zero. Notice also the elliptic estimate, valid without imposing zero average:

$$\|u\|_{H^{s+2}} \leq \|\Delta u\|_{H^s} + \|u\|_{H^s}. \quad (3.11)$$

To prove it, apply the triangle inequality in the weighted coefficient space to $(1 + \lambda_m)a_m = \lambda_m a_m + a_m$. The extra $\|u\|_{H^s}$ controls the constant modes that Δ cannot see. On their orthogonal complement, (3.9) removes that extra term.

Compactness is the uniform control of the high frequencies. Let P_R retain only modes with $\lambda_m \leq R$; its image is finite dimensional. If $\|u\|_{H^2} \leq 1$, then

$$\|u - P_R u\|_{L^2}^2 = \sum_{\lambda_m > R} |\hat{u}(m)|^2 \leq (1 + R)^{-2} \sum_m (1 + \lambda_m)^2 |\hat{u}(m)|^2 \leq (1 + R)^{-2}.$$

The low frequency projections of a bounded H^2 sequence have convergent subsequences in a finite dimensional space. Taking a diagonal subsequence as $R \rightarrow \infty$, the uniform tail estimate

makes that subsequence Cauchy in L^2 . This proves that $H^2 \hookrightarrow L^2$ is compact on the torus. Since $G_0 : L^2 \rightarrow H^2$ is bounded, G_0 is compact when regarded as an operator $L^2 \rightarrow L^2$. There is no contradiction with Example 3.10: a compact operator can have nonclosed image in L^2 . The conclusion we need is that $\Delta : H^2 \rightarrow L^2$ has closed image, the subspace of forms with zero constant coefficient.

Remark 3.14 (What fails on the Euclidean space). On $\mathbb{R}^n$, Fourier frequencies are continuous and can approach zero without being zero. The resulting failure of a uniform gap can be seen without the Fourier transform. Choose a nonzero $\phi \in C_c^\infty(\mathbb{R}^n)$ with $\|\phi\|_{L^2} = 1$ and put $u_R(x) = R^{-n/2}\phi(x/R)$. Then

$$\|u_R\|_{L^2} = 1, \quad \|du_R\|_{L^2} = R^{-1}\|d\phi\|_{L^2}, \quad \|\Delta u_R\|_{L^2} = R^{-2}\|\Delta\phi\|_{L^2}.$$

Thus neither the torus Poincaré inequality nor a uniform estimate $\|u\|_{L^2} \leq C\|\Delta u\|_{L^2}$ holds for all compactly supported smooth functions on $\mathbb{R}^n$. The obstruction is analogous to the eigenvalues $1/j \rightarrow 0$ in Example 3.10: division by arbitrarily small numbers cannot give a bounded inverse. This does not assert that every individual Poisson equation on a noncompact manifold is unsolvable; it identifies the estimate lost when one tries to repeat the compact proof.

3.7 Ellipticity: the highest order part is invertible

Here M may again be any oriented Riemannian manifold without boundary. For a differential operator, use the principal-symbol convention $\partial_{x^j} \mapsto i\xi_j$. At $x \in M$ and $\xi \in T_x^*M$,

$$\sigma_1(d)(x, \xi) = i\xi \wedge, \quad \sigma_1(d^*)(x, \xi) = -i\iota_\xi.$$

The first formula follows from $d(f\alpha) = df \wedge \alpha + f d\alpha$. For the second, the pointwise adjoint of wedge multiplication by ξ is contraction by $\xi^\sharp$; taking the adjoint of i contributes the minus sign. The contraction rule says

$$\iota_{\xi^\sharp}(\xi \wedge \alpha) = |\xi|_g^2 \alpha - \xi \wedge \iota_{\xi^\sharp} \alpha.$$

Consequently,

$$\sigma_2(\Delta)(x, \xi) = \xi \wedge \iota_{\xi^\sharp} + \iota_{\xi^\sharp}(\xi \wedge \cdot) = |\xi|_g^2 \text{id}. \quad (3.12)$$

It is invertible whenever $\xi \neq 0$. This is *ellipticity*. The highest order part controls derivatives in every cotangent direction, even though the full operator can have a kernel. Neither this symbol calculation nor interior elliptic regularity requires compactness or completeness. Locally, if a distributional form u satisfies $\Delta u \in H_{\text{loc}}^s$, then $u \in H_{\text{loc}}^{s+2}$, for every integer $s \geq 0$; the subscript means Sobolev regularity on relatively compact coordinate neighborhoods. In particular, a smooth right-hand side implies a smooth solution. Global Sobolev estimates and compactness of Sobolev inclusions require additional global hypotheses, supplied below by compactness of M .

For the flat torus, (3.12) is exactly the computation (3.5) with $\xi = 2\pi m$, and Fourier modes diagonalize the entire operator. On a general manifold, the metric coefficients vary and there need not be global periodic coordinates. Fourier analysis in one coordinate chart no longer diagonalizes the global problem. Ellipticity retains the invertible highest order part; the following local estimates and compactness theorem replace the explicit frequency calculation.

3.8 The analytic black box, and what we prove from it

For the remainder of this section, assume that M is compact and has no boundary. These hypotheses provide the global analytic estimates needed to pass from the differential identities to existence.

Let $H^s \mathcal{A}^k(M)$ denote the Sobolev completion of smooth forms, using square integrable weak derivatives up to order s for integers $s \geq 0$. Local coordinates, bundle trivializations, and a partition of unity define the norm; different choices give equivalent norms. In particular,

$$\Delta : H^2 \mathcal{A}^k(M) \longrightarrow L^2 \mathcal{A}^k(M)$$

is bounded between these two Hilbert spaces.

Theorem 3.15 (Analytic input for a compact manifold). *Let M be a compact, oriented Riemannian manifold without boundary. For its elliptic operator Δ_k , we use the following three facts.*

1. *There is an elliptic estimate*

$$\|u\|_{H^2} \leq C(\|\Delta u\|_{L^2} + \|u\|_{L^2}), \quad u \in H^2 \mathcal{A}^k(M).$$

2. *The inclusion $H^2 \mathcal{A}^k(M) \hookrightarrow L^2 \mathcal{A}^k(M)$ is compact (Rellich's theorem).*

3. *For each integer $s \geq 0$, if u is a distributional form and $\Delta u \in H^s$, then $u \in H^{s+2}$. In particular, distributional solutions of $\Delta u = 0$ are smooth, and a smooth right-hand side implies a smooth solution.*

The local elliptic estimates and regularity, together with Rellich's compactness theorem, are the analysis deferred in this topic; see [3, lectures 15–16 and 18]. Compactness lets us pass from interior estimates to the global statements above using finitely many coordinate neighborhoods. The local regularity statement remains valid on noncompact manifolds, but does not by itself give a global H^{s+2} bound there. On the flat torus we have already proved all three inputs: the estimate is (3.11), compactness follows from uniformly small high frequency tails, and regularity follows by division by λ_m . The remaining argument explains how the same conclusions follow on a compact manifold without explicit eigenfunctions. We now derive the needed existence theorem, rather than adding it as another black box.

Proposition 3.16 (Finite dimensional kernel and a stronger estimate). *Let M be a compact, oriented Riemannian manifold without boundary. Then the smooth kernel $\mathcal{H}^k(M, g)$ is finite dimensional. There is a constant C_0 such that*

$$\|u\|_{H^2} \leq C_0 \|\Delta u\|_{L^2}, \quad u \in H^2 \mathcal{A}^k(M) \cap (\mathcal{H}^k)^\perp, \quad (3.13)$$

where orthogonality is taken in L^2 .

Proof. Regularity identifies the H^2 kernel with the smooth harmonic forms. If this kernel were infinite dimensional, it would contain an infinite L^2 orthonormal sequence h_j . The elliptic estimate bounds $\|h_j\|_{H^2}$ uniformly. Rellich gives an L^2 convergent subsequence, contradicting $\|h_j - h_\ell\|_{L^2} = \sqrt{2}$ for $j \neq \ell$.

If (3.13) failed, choose $u_j \perp \mathcal{H}^k$ with $\|u_j\|_{H^2} = 1$ and $\|\Delta u_j\|_{L^2} \rightarrow 0$. Rellich gives a subsequence converging in L^2 . The elliptic estimate on $u_j - u_\ell$ then makes that subsequence Cauchy in H^2 , so it converges to u with $\|u\|_{H^2} = 1$. Boundedness of Δ gives $\Delta u = 0$, and L^2 convergence preserves $u \perp \mathcal{H}^k$. Hence $u = 0$, a contradiction. $\square$

On the torus, (3.13) was a direct consequence of $\lambda_m \geq 4\pi^2$ for $m \neq 0$. The contradiction argument above recovers the required bound on a general compact manifold: compactness prevents a normalized sequence in the harmonic complement from becoming arbitrarily close to the kernel under Δ . We do not need to know the eigenfunctions or the numerical value of the spectral gap to use this bound.

Theorem 3.17 (Solving the Laplace equation). *Let M be a compact, oriented Riemannian manifold without boundary. The image of $\Delta : H^2\mathcal{A}^k(M) \rightarrow L^2\mathcal{A}^k(M)$ is closed and equals $(\mathcal{H}^k)^\perp$. For every $f \in L^2\mathcal{A}^k$ orthogonal to $\mathcal{H}^k$, there is a unique $u \in H^2\mathcal{A}^k \cap (\mathcal{H}^k)^\perp$ with $\Delta u = f$. If f is smooth, so is u .*

Proof. Let $\Delta u_j \rightarrow f$ in L^2 . Since $\mathcal{H}^k$ is finite dimensional and smooth, subtracting the L^2 projection of u_j onto $\mathcal{H}^k$ leaves an H^2 form, changes no Δu_j , and makes $u_j \perp \mathcal{H}^k$. Estimate (3.13) now shows that u_j is Cauchy in H^2 . Its limit satisfies $\Delta u = f$, proving closedness.

It remains to identify the orthogonal complement of this image. If $v \in L^2$ is orthogonal to the image, then

$$(v, \Delta\phi)_{L^2} = 0 \quad \text{for every smooth test form } \phi.$$

Formal self-adjointness says exactly that $\Delta v = 0$ in the distributional sense. Weak elliptic regularity makes v smooth, so $v \in \mathcal{H}^k$. Conversely, a smooth harmonic form is orthogonal to the image by integration by parts, first on smooth forms and then by density on H^2 . Thus $(\text{im } \Delta)^\perp = \mathcal{H}^k$. Because the image is closed, the Hilbert space orthogonal-decomposition theorem gives $\text{im } \Delta = (\mathcal{H}^k)^\perp$. Existence follows; two solutions in this complement differ by a harmonic form and hence are equal. Smoothness follows from the regularity black box. $\square$

Definition 3.18 (Harmonic projection and Green operator). Under the compactness and no-boundary hypotheses of Theorem 3.17, let Π_k be the L^2 orthogonal projection onto $\mathcal{H}^k$. Define $G_k f$ to be the unique solution orthogonal to $\mathcal{H}^k$ of

$$\Delta_k G_k f = f - \Pi_k f.$$

Then $G_k : L^2\mathcal{A}^k \rightarrow H^2\mathcal{A}^k$ is bounded by (3.13), vanishes on $\mathcal{H}^k$, and maps smooth forms to smooth forms.

This is the analytic replacement for inverting a nonzero eigenvalue of a finite matrix. Compactness and ellipticity have supplied the closed image and smooth inverse needed for the algebraic argument.

4 Hodge decomposition, duality, and complete examples

The global Hodge theorems below concern compact oriented Riemannian manifolds without boundary. We combine the analytic existence theorem with the algebra of d and d^* . The resulting representative is canonical *after a metric has been fixed*; the underlying cohomology class does not depend on that choice. The pointwise conformal formulas and the local hyperbolic calculations later in this section do not require compactness.

4.1 From the Green operator to the decomposition

Let (M, g) be a compact oriented Riemannian manifold without boundary. On its smooth forms, Definition 3.18 gives

$$\Delta G = G\Delta = I - \Pi, \quad \Pi G = G\Pi = 0. \quad (4.1)$$

The first identity $\Delta G = I - \Pi$ is the defining equation. For the other order, both $G\Delta\alpha$ and $\alpha - \Pi\alpha$ are orthogonal to $\mathcal{H}^k$ and solve the equation $\Delta u = \Delta\alpha$; uniqueness gives equality. The remaining identities follow from the image and kernel conditions on G . Moreover, G is symmetric for the L^2 pairing: for smooth α, β ,

$$(G\alpha, \beta) = (G\alpha, \Delta G\beta) = (\Delta G\alpha, G\beta) = (\alpha, G\beta).$$

The harmonic terms disappear because the image of G is orthogonal to $\mathcal{H}^k$.

Lemma 4.1 (Commutation with the Green operator). *Let (M, g) be a compact oriented Riemannian manifold without boundary, and let G_k be its Green operator on k -forms. On smooth forms, with the degree of each operator understood,*

$$dG_k = G_{k+1}d, \quad d^*G_k = G_{k-1}d^*.$$

Proof. The commutation of d and Δ gives

$$\Delta_{k+1}(dG_k\alpha) = d(\alpha - \Pi_k\alpha) = d\alpha.$$

The exact form $d\alpha$ is orthogonal to harmonic forms. Hence $G_{k+1}d\alpha$ solves the same equation. Both proposed solutions are orthogonal to $\mathcal{H}^{k+1}$: one is exact, and the other lies in the image of G_{k+1} . Uniqueness in Theorem 3.17 proves the first identity. Replace exact by coexact and use $d^*\Delta = \Delta d^*$ to obtain the second identity in exactly the same way. $\square$

Theorem 4.2 (Hodge decomposition). *Let (M, g) be a compact oriented Riemannian n -manifold without boundary. Every smooth k -form has a unique orthogonal decomposition into a harmonic form, an exact form, and a coexact form:*

$$\mathcal{A}^k(M) = \mathcal{H}^k(M, g) \oplus d\mathcal{A}^{k-1}(M) \oplus d^*\mathcal{A}^{k+1}(M). \quad (4.2)$$

More explicitly,

$$\alpha = \Pi\alpha + d(d^*G\alpha) + d^*(dG\alpha). \quad (4.3)$$

Here the spaces outside degrees $0, \dots, n$ are understood to be zero.

Proof. Write $\alpha = \Pi\alpha + \Delta G\alpha$ using (4.1) and expand $\Delta = dd^* + d^*d$. Regularity ensures that all the forms appearing here are smooth. Proposition 3.7 proves pairwise orthogonality, which makes the decomposition a direct sum and makes its three components unique. $\square$

Remark 4.3 (Unique components versus unique potentials). In $\alpha = h + da + d^*b$, the forms h , da , and d^*b are unique. The potentials a and b generally are not: one can replace a by $a + \eta$ with $d\eta = 0$, or b by $b + \zeta$ with $d^*\zeta = 0$. Formula (4.3) supplies distinguished potentials $a = d^*G\alpha$ and $b = dG\alpha$. These are, respectively, coexact and exact. Those additional conditions make the potentials unique within those subspaces: a closed coexact form has squared norm $(d^*\gamma, d^*\gamma) = (\gamma, dd^*\gamma) = 0$, and the dual argument applies to a coclosed exact form.

If $d\alpha = 0$, its coexact component vanishes. Indeed, for $\alpha = h + da + d^*b$, orthogonality gives

$$\|d^*b\|_{L^2}^2 = (\alpha, d^*b)_{L^2} = (d\alpha, b)_{L^2} = 0.$$

Equivalently, $dG\alpha = Gd\alpha = 0$ in (4.3). Thus

$$\ker d = \mathcal{H}^k \oplus d\mathcal{A}^{k-1}(M). \quad (4.4)$$

Theorem 4.4 (Hodge–de Rham theorem). *Let (M, g) be a compact oriented Riemannian manifold without boundary. The map*

$$\mathcal{H}^k(M, g) \longrightarrow H_{\text{dR}}^k(M; \mathbb{R}), \quad h \longmapsto [h]$$

is an isomorphism. Each class has a unique harmonic representative, which is its unique smooth representative of minimal L^2 norm.

Proof. Harmonic forms are closed, so the map is defined. Equation (4.4) proves surjectivity. If a harmonic form is exact, $h = da$, then $\|h\|_{L^2}^2 = (h, da) = (d^*h, a) = 0$, proving injectivity. Finally any representative of $[h]$ is $h + da$, and orthogonality gives

$$\|h + da\|_{L^2}^2 = \|h\|_{L^2}^2 + \|da\|_{L^2}^2.$$

Equality with $\|h\|_{L^2}^2$ occurs precisely when the representative itself equals h . $\square$

The theorem does not change the definition of cohomology. It realizes the quotient $\ker d / \operatorname{im} d$ as the finite dimensional vector space of solutions of a differential equation. In particular,

$$b_k(M) = \dim H_{\mathrm{dR}}^k(M; \mathbb{R}) = \dim \mathcal{H}^k(M, g).$$

4.2 Poincaré duality from a positive integral

Theorem 4.5 (de Rham form of Poincaré duality). *For a compact oriented n -manifold without boundary, the pairing*

$$H_{\mathrm{dR}}^k(M; \mathbb{R}) \times H_{\mathrm{dR}}^{n-k}(M; \mathbb{R}) \longrightarrow \mathbb{R}, \quad ([\alpha], [\beta]) \longmapsto \int_M \alpha \wedge \beta \quad (4.5)$$

is nondegenerate. Equivalently, it identifies $H_{\mathrm{dR}}^{n-k}(M; \mathbb{R})$ with the dual vector space of $H_{\mathrm{dR}}^k(M; \mathbb{R})$.

Proof. If $d\beta = 0$, then $\int_M da \wedge \beta = \int_M d(a \wedge \beta) = 0$ by Stokes. Changing β by an exact form gives zero in the same way, so the pairing is well defined on cohomology.

Take a nonzero class and its nonzero harmonic representative h . Since $*\Delta = \Delta*$, the form $*h$ is harmonic and hence closed. It defines a complementary-degree class, and

$$\int_M h \wedge *h = \|h\|_{L^2}^2 > 0.$$

Thus no nonzero class in the first factor annihilates the second. Interchanging the degrees proves nondegeneracy in the other factor; finite dimensionality turns the pairing into the asserted isomorphism. $\square$

In particular, $b_k = b_{n-k}$. On a connected compact oriented Riemannian manifold without boundary, a top-degree form is harmonic precisely when it is a constant multiple of dV_g : apply $*$ to reduce to harmonic functions. The harmonic representative of the top-degree class with integral c is therefore

$$\frac{c}{\operatorname{Vol}_g(M)} dV_g, \quad \operatorname{Vol}_g(M) = \int_M dV_g. \quad (4.6)$$

The degree-zero and top-degree generators pair to 1 when the latter is normalized to have integral 1.

The pairing (4.5) uses orientation, wedge product, and integration; it uses no metric. The proof chooses a metric to produce a class that pairs nontrivially. Changing the metric changes the harmonic representatives and generally the map $[\alpha] \mapsto [*h_g]$, while leaving the pairing and the Betti numbers unchanged. Also, $*$ need not carry an arbitrary *closed* form to a closed form: the extra coclosed condition is essential. Finally, the wedge of harmonic forms need not be harmonic. A concrete example of each issue appears below.

4.3 The round two-sphere: derivatives, eigenforms, and duality

On the unit sphere, away from the poles, use colatitude θ and longitude ϕ . The outward orientation and round metric give

$$g = d\theta^2 + \sin^2 \theta d\phi^2, \quad dV_g = \sin \theta d\theta \wedge d\phi,$$

$$*d\theta = \sin \theta d\phi, \quad *d\phi = -\frac{1}{\sin \theta} d\theta.$$

These are chart formulas, not globally defined coordinate coframes. In particular, a singular-looking coefficient involving $\sin \theta$ must be interpreted through the underlying global form.

For a smooth function, the Laplacian is

$$\Delta f = -\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) - \frac{1}{\sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}.$$

Take the globally smooth height function $z = \cos \theta$. Then

$$\begin{aligned} dz &= -\sin \theta d\theta, & *dz &= -\sin^2 \theta d\phi, \\ \Delta z &= 2z, & d^*dz &= 2z, \\ d(*dz) &= -2z dV_g, & d^*(*dz) &= 0. \end{aligned}$$

Although dz is closed and exact, $*dz$ is not closed. This directly shows why one cannot prove duality by applying $*$ to an arbitrary closed representative. Commutation gives two eigenforms:

$$\Delta(dz) = 2 dz, \quad \Delta(*dz) = 2 * dz.$$

The second one is coexact, because

$$d^*(z dV_g) = - * dz, \quad dz + *dz = d(z) + d^*(-z dV_g).$$

This is an explicit Hodge decomposition with a nonzero exact part and a nonzero coexact part. Its harmonic part is zero, and $G(dz + *dz) = \frac{1}{2}(dz + *dz)$.

For a numerical check of the energy identity,

$$\int_{S^2} z^2 dV_g = \frac{4\pi}{3}, \quad \int_{S^2} |dz|^2 dV_g = \int_{S^2} \sin^2 \theta dV_g = \frac{8\pi}{3} = 2 \int_{S^2} z^2 dV_g.$$

The cellular cohomology of S^2 has dimensions 1, 0, 1 in degrees 0, 1, 2. Using the de Rham comparison and Hodge theorem therefore gives

$$\mathcal{H}^0(S^2) = \mathbb{R} \cdot 1, \quad \mathcal{H}^1(S^2) = 0, \quad \mathcal{H}^2(S^2) = \mathbb{R} \cdot dV_g.$$

This topological input does not use Poincaré duality. The normalized top representative is $dV_g/(4\pi)$, and its pairing with 1 is 1. More generally, the harmonic projection of $F dV_g$ is

$$\Pi_2(F dV_g) = \left(\frac{1}{4\pi} \int_{S^2} F dV_g \right) dV_g.$$

All three coordinate functions x, y, z have eigenvalue 2, by rotational symmetry or the same calculation after rotating the axis.

4.4 The flat torus: harmonic representatives and all three summands

Start with $T^n = \mathbb{R}^n / \mathbb{Z}^n$, with the standard flat metric and orientation. For an increasing index set $I = (i_1 < \dots < i_k)$ write $dx^I = dx^{i_1} \wedge \dots \wedge dx^{i_k}$. The Fourier analysis in Theorem 3.11 identifies the complete harmonic space:

$$\mathcal{H}^k(T^n, g_0) = \text{span}_{\mathbb{R}}\{dx^I : |I| = k\}, \quad \dim \mathcal{H}^k(T^n, g_0) = \binom{n}{k}. \quad (4.7)$$

Indeed, the Laplacian acts on each coefficient by the scalar flat Laplacian, whose kernel consists of constants. Thus harmonicity forces every coefficient to be constant; conversely every constant form is closed and coclosed. In particular, for any smooth form $\alpha = \sum_I a_I(x) dx^I$,

$$\Pi_k \alpha = \sum_I \bar{a}_I dx^I, \quad \bar{a}_I = \int_{[0,1]^n} a_I(x) dx^1 \cdots dx^n.$$

There is a topological way to read these constants *when α is closed*. Let T_I be a coordinate k -torus, with the coordinates outside I fixed, oriented by the increasing order in I . Moving the fixed coordinates gives homologous subtori, so Stokes' theorem makes $\int_{T_I} \alpha$ independent of those coordinates. Averaging this integral over the complementary coordinates and using Fubini gives

$$\bar{\alpha}_I = \int_{T_I} \alpha, \quad h_{[\alpha]} = \sum_{|I|=k} \left(\int_{T_I} \alpha \right) dx^I. \quad (4.8)$$

For $k = 0$, the statement simply says that a closed function is constant. For $k = n$, it says that the representative is its total integral times $dx^1 \wedge \cdots \wedge dx^n$. Thus periods recover every harmonic representative, and a closed form is exact if and only if all its coordinate periods vanish. For forms that are not closed, the coefficient averages still give $\Pi\alpha$, but integrals along individual coordinate subtori need not give those averages.

Specialize to $T^2 = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$, with $z = x + iy$ and $g_0 = dx^2 + dy^2$. The one-forms dx, dy descend from the plane even though x, y are not real-valued global functions on the torus. Equation (4.7) becomes

$$\mathcal{H}^0 = \mathbb{R}, \quad \mathcal{H}^1 = \mathbb{R} dx \oplus \mathbb{R} dy, \quad \mathcal{H}^2 = \mathbb{R} (dx \wedge dy).$$

The star sends 1 to $dx \wedge dy$, dx to dy , and dy to $-dx$; each image is again harmonic. Constant forms also wedge to constant forms. This last property is special to the flat metric and does not hold for arbitrary metrics, even on the same torus.

Example 4.6 (A closed representative and its energy). For any real ε , consider

$$\alpha_\varepsilon = (1 + 2\pi\varepsilon \cos(2\pi x))dx = dx + d(\varepsilon \sin(2\pi x)).$$

Its horizontal and vertical periods are 1 and 0, respectively. Its harmonic representative is therefore dx , and direct integration checks the minimizing property:

$$\|\alpha_\varepsilon\|_{L^2}^2 = 1 + 2\pi^2\varepsilon^2, \quad \|dx\|_{L^2}^2 = 1.$$

For $\varepsilon \neq 0$, the representative is closed but not harmonic: $d^*\alpha_\varepsilon = 4\pi^2\varepsilon \sin(2\pi x) \neq 0$. The nonzero horizontal period prevents dx itself from having a global periodic primitive.

Example 4.7 (Solving the decomposition for an arbitrary one-form). Let $\alpha = P dx + Q dy$, where P, Q are smooth and periodic. We seek

$$\alpha = h + df + d^*(\psi dx \wedge dy), \quad h = \bar{P} dx + \bar{Q} dy, \quad \int_{T^2} f = \int_{T^2} \psi = 0. \quad (4.9)$$

Since $d^*(\psi dx \wedge dy) = \psi_y dx - \psi_x dy$, the required coefficient equations are

$$P - \bar{P} = f_x + \psi_y, \quad Q - \bar{Q} = f_y - \psi_x.$$

Taking divergence and curl separates the unknowns:

$$\Delta f = -(P_x + Q_y), \quad \Delta \psi = Q_x - P_y, \quad \Delta = -\partial_x^2 - \partial_y^2. \quad (4.10)$$

This step also explains the solvability condition: the integral of each right-hand side is zero because the boundary values match on opposite sides of a fundamental square. Each equation therefore has a unique smooth mean-zero solution by the Fourier construction in Section 3.6. With Fourier modes $e^{2\pi i(mx+ny)}$, the solutions for $(m, n) \neq (0, 0)$ are

$$\begin{aligned} \hat{f}(m, n) &= -\frac{i(m\hat{P}(m, n) + n\hat{Q}(m, n))}{2\pi(m^2 + n^2)}, \\ \hat{\psi}(m, n) &= \frac{i(m\hat{Q}(m, n) - n\hat{P}(m, n))}{2\pi(m^2 + n^2)}. \end{aligned}$$

Set $\hat{f}(0,0) = \hat{\psi}(0,0) = 0$ to impose the normalization. Smoothness follows from rapid decay of the input coefficients; division by $m^2 + n^2$ compensates for the derivative in the numerator. To verify that the two Poisson equations suffice, subtract the proposed right-hand side of (4.9) from α . The remainder is closed and coclosed by (4.10), so its coefficients are constant. Their averages are zero, hence the remainder vanishes. The mean-zero normalization removes the constant ambiguity in both potentials.

If α is closed, then $Q_x - P_y = 0$, so $\psi = 0$ and $\alpha = h + df$. Only in this closed case can the coefficients of h always be read as the horizontal and vertical periods. For example, $\cos(2\pi y)dx$ has zero harmonic projection but has horizontal integral $\cos(2\pi y_0)$ along the loop at height y_0 .

For a fully numerical instance of all three summands, set $f = \sin(2\pi x)$, $\beta = \cos(2\pi y) dx \wedge dy$, and consider

$$\begin{aligned}\alpha &= dx + 2dy + df + d^*\beta \\ &= (1 + 2\pi \cos(2\pi x) - 2\pi \sin(2\pi y))dx + 2dy.\end{aligned}$$

The exact, coexact, and harmonic parts are respectively

$$df = 2\pi \cos(2\pi x) dx, \quad d^*\beta = -2\pi \sin(2\pi y) dx, \quad h = dx + 2dy.$$

Their orthogonality is visible by integrating the sine and cosine terms over the unit square. Specifically,

$$\|h\|_{L^2}^2 = 5, \quad \|df\|_{L^2}^2 = 2\pi^2, \quad \|d^*\beta\|_{L^2}^2 = 2\pi^2, \quad \|\alpha\|_{L^2}^2 = 5 + 4\pi^2.$$

Both nonconstant terms have eigenvalue $4\pi^2$, so

$$G\alpha = \frac{1}{4\pi^2}(df + d^*\beta), \quad d^*G\alpha = f, \quad dG\alpha = \beta.$$

The last identity uses $d\beta = 0$ and $dd^*\beta = \Delta\beta = 4\pi^2\beta$; similarly $d^*df = 4\pi^2f$. Substituting in (4.3) recovers the original decomposition exactly.

The period-normalized basis of H^1 is $[dx], [dy]$, and

$$\int_{T^2} dx \wedge dy = 1, \quad \left(\int_{T^2} e_i \wedge e_j \right)_{i,j} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (e_1, e_2) = (dx, dy).$$

This is the nondegenerate alternating Poincaré pairing in degree one.

4.5 Flat metrics, periods, and the four-dimensional pairing

On the same smooth unit torus put $g_{A,B} = A^2 dx^2 + B^2 dy^2$, with $A, B > 0$. Its scalar Laplacian is $-A^{-2}\partial_x^2 - B^{-2}\partial_y^2$; all nonzero Fourier modes still have positive eigenvalues. Thus the harmonic spaces in every degree remain the constant forms. The representatives of $[dx]$, $[dy]$, and the integral-one top class remain dx , dy , and $dx \wedge dy$. But orthonormalizing the coframe gives

$$*dx = \frac{B}{A} dy, \quad *dy = -\frac{A}{B} dx, \quad dV_g = AB dx \wedge dy,$$

and consequently

$$((e_i, e_j)_{L^2})_{i,j} = \begin{pmatrix} B/A & 0 \\ 0 & A/B \end{pmatrix}, \quad (e_1, e_2) = (dx, dy).$$

The Poincaré matrix remains $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. This example separates three objects: a harmonic representative, the metric inner product on the harmonic space, and the integration pairing on cohomology. The first happens to stay fixed throughout this family; the second varies; the third is topological.

The same distinction is visible for an oblique complex torus $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, with $\tau = a + ib$, $b > 0$. Using $z = x + \tau y$ pulls its Euclidean metric back to

$$g_\tau = (dx + a dy)^2 + b^2 dy^2, \quad dV_{g_\tau} = b dx \wedge dy.$$

The metric coefficients are constant and positive definite, so again all harmonic forms have constant coefficients. On harmonic one-forms the two matrices are

$$((e_i, e_j)_{L^2})_{i,j} = \frac{1}{b} \begin{pmatrix} a^2 + b^2 & -a \\ -a & 1 \end{pmatrix}, \quad \left(\int_{T^2} e_i \wedge e_j \right)_{i,j} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For instance, the star is $*dx = (a/b)dx + ((a^2 + b^2)/b)dy$ and $*dy = -(1/b)dx - (a/b)dy$; inserting these into $\int \alpha \wedge *\beta$ recovers the first matrix. The integral-one harmonic top form is $dV_g/b = dx \wedge dy$.

In four dimensions, take the standard oriented flat T^4 and write $e^{ij} = dx^i \wedge dx^j$. All six e^{ij} are harmonic. For the intersection pairing $Q(\alpha, \beta) = \int_{T^4} \alpha \wedge \beta$, use the three ordered pairs

$$(e^{12}, e^{34}), \quad (e^{13}, -e^{24}), \quad (e^{14}, e^{23}).$$

Within each pair (u, v) we have $u \wedge v = dV$ and $u \wedge u = v \wedge v = 0$; products between distinct pairs vanish because an index is repeated. Hence, in this ordered basis,

$$Q = U \oplus U \oplus U, \quad U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (4.11)$$

The star exchanges u and v in each pair. The six forms $(u + v)/\sqrt{2}$ and $(u - v)/\sqrt{2}$ therefore give orthonormal bases of the self-dual and anti-self-dual harmonic spaces, respectively:

$$\dim \mathcal{H}_+^2(T^4) = \dim \mathcal{H}_-^2(T^4) = 3, \quad Q|_{\mathcal{H}_+^2} > 0, \quad Q|_{\mathcal{H}_-^2} < 0.$$

Here the signs follow directly from $\int \alpha \wedge \alpha = \pm \int \alpha \wedge *\alpha = \pm \|\alpha\|_{L^2}^2$ on the two eigenspaces. The pairing is *symmetric* in degree two because $(-1)^{2 \cdot 2} = 1$; this symmetry of Q is a different notion from self-duality of an individual form.

4.6 Changing the metric: a surface phenomenon

For a conformal change $\tilde{g} = e^{2u}g$ on any oriented Riemannian n -manifold, compactness is unnecessary: the pointwise inner product on k -forms scales by e^{-2ku} and the volume form by e^{nu} . The defining identity for the star gives

$$*_\tilde{g}|_{\mathcal{A}^k} = e^{(n-2k)u} *_g|_{\mathcal{A}^k}. \quad (4.12)$$

On a surface, $*$ on one-forms is therefore conformally invariant. For a one-form α , this implies

$$d_\tilde{g}^* \alpha = e^{-2u} d_g^* \alpha.$$

The conditions $d\alpha = 0$ and $d^*\alpha = 0$ are unchanged on every oriented surface. If the surface is compact and without boundary, these conditions are equivalent to $\Delta\alpha = 0$. Hence its harmonic one-forms, and the harmonic representatives of fixed cohomology classes, depend only on the conformal class of the metric. This is a special feature of the middle degree; it does not assert that all harmonic representatives are metric independent.

Remark 4.8 (The full smooth kernel on a noncompact surface). The compactness qualification cannot be dropped when harmonic means $\Delta\alpha = 0$. On the Euclidean plane, $\alpha = x dx$ satisfies

$$\Delta_{g_0}\alpha = 0, \quad d\alpha = 0, \quad d_{g_0}^*\alpha = -1.$$

For $\tilde{g} = e^{2x}g_0$, the preceding formula gives $d_{\tilde{g}}^*\alpha = -e^{-2x}$, and consequently

$$\Delta_{\tilde{g}}\alpha = d(-e^{-2x}) = 2e^{-2x}dx \neq 0.$$

Thus the full smooth kernel of Δ need not be conformally invariant on a noncompact surface, although the space of closed and coclosed one-forms is.

On the unit torus let $g_u = e^{2u(x,y)}(dx^2 + dy^2)$ with u smooth and periodic. The complete harmonic spaces are

$$\begin{aligned} \mathcal{H}^0(T^2, g_u) &= \mathbb{R}, & \mathcal{H}^1(T^2, g_u) &= \mathbb{R} dx \oplus \mathbb{R} dy, \\ \mathcal{H}^2(T^2, g_u) &= \mathbb{R} e^{2u} dx \wedge dy. \end{aligned}$$

The representatives of $[dx]$ and $[dy]$ remain dx and dy . Indeed, their stars are still dy and $-dx$, so they are both closed and coclosed. The integral-one top-degree class has representative

$$\frac{e^{2u(x,y)}}{\int_{T^2} e^{2u} dx dy} dx \wedge dy.$$

If u is nonconstant, this differs from $dx \wedge dy$. In fact

$$*_{g_u}(dx \wedge dy) = e^{-2u}$$

is nonconstant, so $dx \wedge dy$ is not harmonic for g_u . Thus dx and dy are harmonic while their wedge is not. The ring structure on cohomology is computed by wedging closed representatives and, if desired, projecting the result back to harmonic forms. For example, this projection sends $dx \wedge dy$ to the normalized volume form just displayed: the integral fixes its cohomology class.

Example 4.9 (A metric that changes a harmonic one-form). To see an actual change of degree-one representatives, choose a smooth positive 1-periodic function $\rho(x)$ and put

$$g_\rho = \rho(x)^2 dx^2 + dy^2, \quad L = \int_0^1 \rho(x) dx, \quad dV_{g_\rho} = \rho(x) dx \wedge dy.$$

This changes the two coordinate directions differently. Its star is

$$*dx = \rho^{-1}dy, \quad *dy = -\rho dx, \quad *(\rho dx \wedge dy) = 1.$$

We can find the representative of $[dx]$ by solving its coclosed condition, rather than guessing. Seek $h_x = dx + df$ with $f = f(x)$ periodic. Since this is already closed, harmonicity is equivalent to

$$0 = d(*h_x) = \frac{d}{dx} \left(\frac{1 + f'(x)}{\rho(x)} \right) dx \wedge dy.$$

Thus $1 + f' = C\rho$. Integrating over a period forces $1 = CL$, which is exactly the compatibility condition for a periodic primitive. We obtain

$$h_x = \frac{\rho(x)}{L} dx, \quad h_y = dy, \quad h_2 = \frac{\rho(x)}{L} dx \wedge dy. \quad (4.13)$$

All are closed, and $*h_x = dy/L$, $*h_y = -\rho dx$, and $*h_2 = 1/L$ are closed as well. They are therefore harmonic. Their coordinate periods, or integral in top degree, are normalized to 1 in the indicated direction. They give the full spaces

$$\mathcal{H}^0(T^2, g_\rho) = \mathbb{R}, \quad \mathcal{H}^1(T^2, g_\rho) = \mathbb{R}h_x \oplus \mathbb{R}h_y, \quad \mathcal{H}^2(T^2, g_\rho) = \mathbb{R}h_2.$$

One direct completeness check is the global change of circle coordinate $s = L^{-1} \int_0^x \rho(t) dt \pmod{1}$. It gives $ds = h_x$ and $g_\rho = L^2 ds^2 + dy^2$, reducing the question to the rectangular torus.

For a particularly explicit choice, take $\rho(x) = 1 + \varepsilon \cos(2\pi x)$ with $|\varepsilon| < 1$. Then $L = 1$, and

$$h_x = (1 + \varepsilon \cos(2\pi x))dx = dx + d\left(\frac{\varepsilon}{2\pi} \sin(2\pi x)\right).$$

The form dx is the harmonic representative for g_0 , while h_x is the harmonic representative of the same class for g_ρ . When $\varepsilon \neq 0$, $d(*_{g_\rho} dx) = d(\rho^{-1} dy) \neq 0$, so dx has lost harmonicity. This does not contradict conformal invariance: g_ρ is not a conformal rescaling of g_0 in the fixed coordinates.

Finally,

$$\int_{T^2} h_x \wedge h_y = 1, \quad \|h_x\|_{L^2(g_\rho)}^2 = \frac{1}{L}, \quad \|h_y\|_{L^2(g_\rho)}^2 = L, \quad (h_x, h_y)_{L^2(g_\rho)} = 0.$$

The period pairing stays fixed while the actual representatives and their metric lengths are determined by the metric. The calculation $1 + f' = C\rho$ is a one-dimensional model of the general analytic strategy: impose the desired differential equation, identify the integral obstruction, and normalize the remaining kernel ambiguity.

4.7 The hyperbolic upper half-plane and a compact quotient

The following differential-operator formulas are local and require no compactness assumption. On $\mathbb{H} = \{x + iy : y > 0\}$ take

$$g_{\text{hyp}} = \frac{dx^2 + dy^2}{y^2}, \quad dV_{\text{hyp}} = \frac{dx \wedge dy}{y^2}.$$

The oriented orthonormal coframe is $dx/y, dy/y$. Consequently

$$*1 = \frac{dx \wedge dy}{y^2}, \quad *dx = dy, \quad *dy = -dx, \quad *(dx \wedge dy) = y^2.$$

For $\alpha = P dx + Q dy$, the full computation is

$$*\alpha = P dy - Q dx, \quad d(*\alpha) = (P_x + Q_y)dx \wedge dy, \quad d^*\alpha = -y^2(P_x + Q_y).$$

In particular, with our nonnegative convention,

$$\Delta f = -y^2(f_{xx} + f_{yy}). \tag{4.14}$$

For a top form $R dV_{\text{hyp}}$, one also obtains

$$d^*(R dV_{\text{hyp}}) = R_y dx - R_x dy.$$

Now let $\Gamma \subset \text{PSL}_2(\mathbb{R})$ be a torsion-free cocompact discrete subgroup and let $\Sigma = \Gamma \backslash \mathbb{H}$. The action is by orientation-preserving hyperbolic isometries, so the metric, volume form, and differential operators descend to a smooth compact oriented surface. Write its genus as $g \geq 2$. The forms dx and dy themselves generally do *not* descend. To construct global examples, require invariance under the group.

Let $F(z) dz$ be a holomorphic differential on $\mathbb{H}$ satisfying

$$F(\gamma z) \gamma'(z) = F(z) \quad (\gamma \in \Gamma). \tag{4.15}$$

This is exactly the condition for the differential to descend to Σ . Write $F = P + iQ$. Its real and imaginary parts are

$$\alpha = P dx - Q dy, \quad *\alpha = Q dx + P dy.$$

The Cauchy–Riemann equations $P_x = Q_y$ and $P_y = -Q_x$ imply

$$d\alpha = -(Q_x + P_y)dx \wedge dy = 0, \quad d*\alpha = -y^2(P_x - Q_y) = 0.$$

Thus α and $*\alpha$ are harmonic downstairs; this direction uses only closedness and coclosedness, not compactness. Conversely, if $\alpha = A dx + B dy$ is a harmonic real one-form lifted from the compact surface Σ , the energy identity on Σ makes it closed and coclosed. Thus its coefficients satisfy

$$B_x - A_y = 0, \quad A_x + B_y = 0.$$

These are the Cauchy–Riemann equations for $A - iB$. Hence $\alpha + i*\alpha = (A - iB)dz$ is an invariant holomorphic differential. This provides an explicit local dictionary between harmonic real one-forms and holomorphic differentials. Finding the invariant function F for a specified group is an additional global problem; the local formula alone does not produce a differential on the quotient.

The usual cellular calculation for a closed genus- q surface gives $b_0 = b_2 = 1$ and $b_1 = 2q$. The Hodge theorem therefore yields

$$\mathcal{H}^0(\Sigma) = \mathbb{R}, \quad \dim_{\mathbb{R}} \mathcal{H}^1(\Sigma) = 2q, \quad \mathcal{H}^2(\Sigma) = \mathbb{R} dV_{\text{hyp}}.$$

The displayed local dictionary identifies $\mathcal{H}^1(\Sigma)$ with the real vector space underlying the q -dimensional complex space of holomorphic differentials. For any nonzero such real form,

$$\int_{\Sigma} \alpha \wedge *\alpha = \int_{\mathcal{F}} (P^2 + Q^2) dx dy > 0,$$

where $\mathcal{F}$ is a fundamental domain; the integrand is the pullback of the globally defined two-form $\alpha \wedge *\alpha$. This is Poincaré duality expressed in hyperbolic coordinates. If one uses Gauss–Bonnet, constant curvature -1 gives $\text{Area}(\Sigma) = 4\pi(q - 1)$, so its integral-one top representative is $dV_{\text{hyp}}/[4\pi(q - 1)]$.

Remark 4.10 (Why the upper half-plane itself is different). On the noncompact space $\mathbb{H}$, the function $f(x, y) = y$ satisfies $\Delta f = 0$ by (4.14), while $df = dy \neq 0$. There is no contradiction with (3.3): neither f nor df has finite L^2 norm on all of $\mathbb{H}$, and a global integration by parts cannot discard the behavior at infinity. The function y also does not descend to a compact quotient. Compactness and the absence of boundary are substantive hypotheses of the Hodge theorem proved here.

5 Exercises and hints

Exercises 1–5 review the topological and pointwise constructions; 6–10 develop the operator and finite-dimensional arguments; 11–16 connect geometry with global cohomology. Exercises 17–22 use tori to work through the metric, harmonic representatives, Poisson equations, convergence, and dependence of estimates on geometry. Detailed solutions are in the blackboard companion.

Exercise 1 (A period detects a hole). On $U = \mathbb{R}^2 \setminus \{0\}$ let

$$\eta = \frac{-y dx + x dy}{2\pi(x^2 + y^2)}.$$

Compute $d\eta$ and its integral around the positively oriented unit circle. Compute $H_{\text{dR}}^k(U)$ in all degrees. Show that $\eta + d(x/(x^2 + y^2))$ is a different representative of the same class.

Hint. Retract U onto the unit circle; on the circle subtract the average of a periodic coefficient before integrating.

Exercise 2 (The radial primitive). Using the homotopy $F(t, x, y) = (tx, ty)$, compute $K\alpha$ for

$$\alpha = x dx + y dy, \quad \text{and } K\omega \text{ for } \omega = dx \wedge dy.$$

Verify the primitive identities directly. Explain why the same proof cannot contract the punctured plane to the origin inside itself.

Hint. $K\omega = \frac{1}{2}(x dy - y dx)$; check the factor $1/2$ by differentiation.

Exercise 3 (Periods on the square torus). Let $\alpha = P dx + Q dy$ be closed on T^2 . Let a, b be its periods around the horizontal and vertical circles. Prove $\alpha = a dx + b dy + df$ with f periodic, unique up to a constant. Compute the matrix of the integration pairing on H^1 in the basis $[dx], [dy]$.

Hint. Pull back to $\mathbb{R}^2$, take a primitive, and use the two zero periods to prove its translation invariance.

Exercise 4 (Use the inverse metric). On $\mathbb{R}^2$ with orientation $dx \wedge dy$ and metric $g = 4 dx^2 + 9 dy^2$, compute $|dx|_g^2$, $|dy|_g^2$, dV_g , $*dx$, $*dy$, and $*(dx \wedge dy)$. Compute $\|dx + dy\|_{L^2([0,1]^2, g)}^2$.

Hint. The orthonormal coframe is $2dx, 3dy$; the squared norm of a covector uses g^{-1} .

Exercise 5 (Four-dimensional splitting). With the standard orientation and metric on $\mathbb{R}^4$, let

$$\alpha = dx^1 \wedge dx^2 + 2 dx^3 \wedge dx^4 + dx^1 \wedge dx^3.$$

Compute $*\alpha$, $\alpha_{\pm} = \frac{1}{2}(\alpha \pm *\alpha)$, and $\alpha \wedge \alpha$. Verify the identity involving $|\alpha_+|^2 - |\alpha_-|^2$. Explain why α_+ is still an alternating tensor.

Hint. $*(dx^1 \wedge dx^3) = -dx^2 \wedge dx^4$.

Exercise 6 (Three-dimensional vector calculus). For $v = (P, Q, R)$ on Euclidean $\mathbb{R}^3$, write $v^\flat = Pdx + Qdy + Rdz$. Compute $*d(v^\flat)$ and $d^*(v^\flat)$, and identify curl and minus divergence. Evaluate both for $v = (-y, x, 0)$.

Hint. $*(dx \wedge dy) = dz$, $*(dy \wedge dz) = dx$, $*(dz \wedge dx) = dy$.

Exercise 7 (Signs in the plane). For $\alpha = Pdx + Qdy$ and $\beta = Rdx \wedge dy$ on Euclidean $\mathbb{R}^2$, derive

$$d^*\alpha = -(P_x + Q_y), \quad d^*\beta = R_y dx - R_x dy.$$

Compute $\Delta\alpha$ by expanding both terms and canceling mixed derivatives. Test on $\alpha = x^2dx + xy dy$.

Hint. On an oriented Riemannian surface $d^* = -*d*$ in every degree in which the formula is nonzero.

Exercise 8 (A complete finite-dimensional model). Give $C^0 = \mathbb{R}$, $C^1 = \mathbb{R}^3$, $C^2 = \mathbb{R}$ their standard inner products, with

$$d_0(t) = (t, 0, 0), \quad d_1(a, b, c) = c.$$

Write the matrices for both adjoints and for Δ_1 . Find all three Hodge summands, the cohomology in each degree, and the harmonic representative of a degree-one class.

Hint. The three coordinate axes in C^1 have different roles.

Exercise 9 (Why closure matters). Let $A : \ell^2 \rightarrow \ell^2$ be $(Ax)_j = x_j/j$. Prove that $\text{im } A$ is dense and not closed. Show that $y = (1/j)_{j \geq 1}$ belongs to its closure but not its image. Explain why $(\text{im } A)^\perp = \ker A^*$ does not give an algebraic decomposition $\ell^2 = \ker A^* \oplus \text{im } A$.

Hint. Every finitely supported sequence is in the image, whereas a preimage of y would have every entry equal to 1.

Exercise 10 (Harmonic representatives minimize norm). On a compact oriented Riemannian manifold without boundary, prove directly that an exact harmonic form vanishes. Then show that a harmonic representative h has smaller L^2 norm than every distinct representative of $[h]$.

Hint. Use $\langle h, d\beta \rangle_{L^2} = \langle d^*h, \beta \rangle_{L^2} = 0$, followed by the Pythagorean identity.

Exercise 11 (A sphere calculation that is global). On the unit round S^2 , use polar coordinates away from the poles and put $z = \cos \theta$. Compute dz , $*dz$, d^*dz , $d(*dz)$ and Δz . Show that $*dz$ is smooth at the poles using ambient coordinates (x, y, z) . Compute $\|dz\|_{L^2}^2$ and verify the energy identity.

Hint. $x dy - y dx = \sin^2 \theta d\phi$ on S^2 , and $\int_{S^2} z^2 dV = 4\pi/3$.

Exercise 12 (An explicit Hodge decomposition). On the flat unit torus, decompose

$$\alpha = (1 + 2\pi \cos(2\pi x) - 2\pi \sin(2\pi y))dx + 2dy$$

into harmonic, exact, and coexact components, displaying potentials for the last two. Compute $d\alpha$, $d^*\alpha$, and the three squared L^2 norms. Is $[\alpha]$ a de Rham cohomology class?

Hint. Use $f = \sin(2\pi x)$ and $\beta = \cos(2\pi y)dx \wedge dy$. A Hodge decomposition exists even when the original form is not closed.

Exercise 13 (Conformal invariance has a degree). For $g' = e^{2u}g$ on an oriented n -manifold, derive

$$*_{g'}|_{\mathcal{A}^k} = e^{(n-2k)u} *_{g}.$$

On a closed oriented surface, show that the spaces of harmonic one-forms for g and g' coincide. On the square torus with $g' = e^{2u}(dx^2 + dy^2)$, find the harmonic representative of the class whose integral is 1 in degree two.

Hint. The criterion is closed and coclosed; the normalized top form is $dV_{g'}/\int dV_{g'}$.

Exercise 14 (A hyperbolic quotient). On $\mathbb{H} = \{y > 0\}$ with $g = y^{-2}(dx^2 + dy^2)$, compute $d^*(Pdx + Qdy)$ and Δf . If $F = P + iQ$ is holomorphic, prove that $\operatorname{Re}(Fdz) = Pdx - Qdy$ is closed and coclosed. Let $\Gamma \subset \operatorname{PSL}_2(\mathbb{R})$ be discrete and torsion-free. State the condition under which this form descends to $\Gamma \backslash \mathbb{H}$; cocompactness is not required for descent or for the closed and coclosed equations. Finally compute Δy and dy , and explain why there is no contradiction with the compact energy argument.

Hint. Descent means $F(\gamma z)\gamma'(z) = F(z)$; local coordinates themselves need not descend.

Exercise 15 (Two dualities, two signs). Let M be a compact oriented n -manifold without boundary. Prove $Q(\beta, \alpha) = (-1)^{k(n-k)}Q(\alpha, \beta)$ for its integration pairing in complementary degrees. Explain why the middle pairing is skew-symmetric in dimension two but symmetric in dimension four. If $n = 4$, choose a Riemannian metric and prove that Q is positive definite on harmonic self-dual two-forms and negative definite on harmonic anti-self-dual two-forms.

Hint. Integrate $\alpha \wedge \alpha = \pm |\alpha|^2 dV$ on the two eigenspaces of $*$.

Exercise 16 (A metric changes the representative). On $S^1 = \mathbb{R}/\mathbb{Z}$, take $g = a(t)^2 dt^2$ with $a > 0$. Find the harmonic one-form with period 1. Specialize to $a(t) = 1 + \varepsilon \cos(2\pi t)$, $|\varepsilon| < 1$, and compare it with dt . What additional structure made dt a distinguished representative initially?

Hint. $*(b(t)dt) = b/a$, so coclosedness means $(b/a)' = 0$.

Exercise 17 (The oblique torus). Let $T_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, with $\tau = a + ib$, $b > 0$, and write $z = x + \tau y$ with x, y of period one. For the metric $g = |dz|^2$, compute the metric matrix, area, $*dx$, $*dy$, and the L^2 Gram matrix of dx, dy . Show that $*dz = -i dz$ and $\|dz\|_{L^2}^2 = 2b$. Give period-normalized harmonic representatives in every degree and compare their metric pairing with their Poincaré pairing.

Hint. The determinant of the metric is b^2 . Constant coefficients imply both closedness and coclosedness for this constant metric.

Exercise 18 (The torus Poisson equation). On the unit flat T^2 , prove that $\Delta u = f$ has a smooth solution for smooth f if and only if $\int_{T^2} f = 0$. Describe all solutions and compute the mean-zero one for $f = \sin(2\pi x) + \cos(4\pi y)$. For general smooth f , explain why dividing its nonzero Fourier coefficients by $4\pi^2(m^2 + n^2)$ produces a smooth function. Prove the estimate $\|u\|_{L^2} \leq (4\pi^2)^{-1} \|f\|_{L^2}$ for the mean-zero solution.

Hint. Separate the zero mode, use Parseval, and use rapid Fourier decay to justify termwise differentiation of the solution series.

Exercise 19 (One frequency, two projections). On the unit flat T^2 , decompose $\alpha = \cos(2\pi(x + y))dx$ into harmonic, exact, and coexact components. Find mean-zero potentials v, ψ such that the last two components are dv and $d^*(\psi dx \wedge dy)$. Compute their squared norms and verify orthogonality both pointwise and after integration. Does this form define a cohomology class?

Hint. Project the coefficient vector $(1, 0)$ onto $(1, 1)$ and $(1, -1)$. The two potentials can be chosen equal.

Exercise 20 (A changing representative on the same torus). Take $g_\rho = \rho(x)^2 dx^2 + dy^2$ on $\mathbb{R}^2/\mathbb{Z}^2$, with $\rho > 0$ periodic and $L = \int_0^1 \rho$. Find the harmonic representatives h_x, h_y, h_2 of $[dx], [dy], [dx \wedge dy]$ and their L^2 norms. Prove completeness of your list using the circle coordinate $X = L^{-1} \int_0^x \rho(t) dt$ modulo 1. For $\rho = 1 + \varepsilon \cos(2\pi x)$, $|\varepsilon| < 1$, exhibit explicitly the exact difference $h_x - dx$.

Hint. The metric becomes $L^2 dX^2 + dy^2$. The coordinate X is circle-valued; it is not a globally defined real primitive for h_x .

Exercise 21 (The intersection pairing of the four-torus). On the unit flat T^4 , let $e^{ij} = dx^i \wedge dx^j$, with orientation $dx^1 \wedge dx^2 \wedge dx^3 \wedge dx^4$. Show that the six forms e^{ij} are a basis of harmonic two-forms. In the ordered pairs (e^{12}, e^{34}) , $(e^{13}, -e^{24})$, (e^{14}, e^{23}) , compute the three blocks of the integration pairing. Find b_2^+ and b_2^- . For $\omega = e^{12} + e^{34}$, compute $*\omega$, $\|\omega\|_{L^2}^2$, and $\int \omega^2$.

Hint. Each pair contributes the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Distinguish harmonic forms from arbitrary smooth self-dual sections.

Exercise 22 (An inverse estimate depends on the metric). Give T^2 the metric $g_L = L^2 dx^2 + dy^2$, with $L \geq 1$. Compute the eigenvalue of $e^{2\pi i(mx+ny)}$, the first positive eigenvalue, and the norm of the Green operator on the mean-zero L^2 subspace. Solve $\Delta_L u = \sin(2\pi x)$ with zero mean. Show that a single bound $\|u\|_{L^2(g_L)} \leq C \|\Delta_L u\|_{L^2(g_L)}$ cannot hold for all $L \geq 1$ with fixed C .

Hint. Every fixed torus is compact; the loss of a uniform bound occurs as the metric changes and the first positive eigenvalue tends to zero.

References

- [1] The Stacks Project Authors, *The Stacks Project*, Derived Categories, Lemma 13.16.7 (Leray’s acyclicity lemma), Tag 015E. <https://stacks.math.columbia.edu/tag/015E>. Used for the acyclic-resolution theorem in Section 1.3.
- [2] Frank W. Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Graduate Texts in Mathematics 94, Springer, 1983, Chapter 5, “The de Rham Theorem,” pp. 161–217. https://doi.org/10.1007/978-1-4757-1799-0_5. Provides the sheaf-cohomology proof, comparison with differentiable and continuous singular cohomology, and identification of the ring isomorphism by integration.
- [3] Semyon Dyatlov, *Differential Analysis I*, MIT 18.155, Fall 2021. Lectures 15–16: elliptic regularity, estimates, compactness, and Fredholm theory; Lecture 18: vector bundles, Hodge decomposition, and Poincaré duality. <https://math.mit.edu/~dyatlov/18.155-F21/>. In particular, see [Lecture 15](#), [Lecture 16](#), and [Lecture 18](#).
- [4] R. O. Wells, Jr., *Differential Analysis on Complex Manifolds*. Further reading on elliptic operators and Hodge theory, also recommended in the full course notes.